

# TP LAB 1 – Singlestage

Applied Electronics 1 (R3571) – UTN FRBA – 2024

**Revision:** 2.0

**Date:** 07/10/2024

**Group:**7

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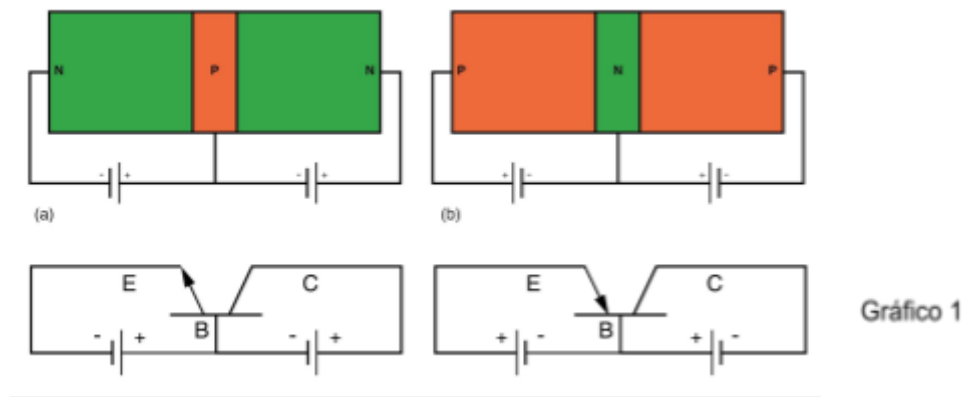
## 1. Introduction

### Goals

- Apply calculation methods for the analysis and design of single-stage amplifier configurations.
- Verify the practical functioning of the designs.
- Get hands-on with real electronic devices.
- Acquire skill in assembling and starting up a prototype.
- To familiarize ourselves with the use of laboratory equipment.
- To be able to resolve the problems that arise when starting up a circuit.
- Compare the measurements taken with the theoretical calculations.
- Practice using simulation applications such as LTspice.

### TBJ

The bipolar junction transistor is a semiconductor device consisting of two P-N junctions that can be used both as a "switch" and as an amplifier; there are two types depending on the combination of zones, as can be seen in Figure 1.



*P-N junctions for NPN (a) and PNP (b) bipolar transistors.*

Its operation is based on the control of the flow of current through the junctions (electrons or holes).

It has three working zones: cutoff, where the transistor is "off", saturation, where it allows current to pass through, and the active zone, where it functions as an amplifier.

For a BJT to operate within this region and achieve current flow between the emitter and collector based on the small base current, proper biasing of both junctions is necessary. Forward bias is used for the emitter-base junction, while reverse bias is used for the collector-base junction.

Ideally, the collector current is a function of the base current; however, in reality, there is a dependence on  $V_{ce}$ . The following formulas express the current relationships.

DONE

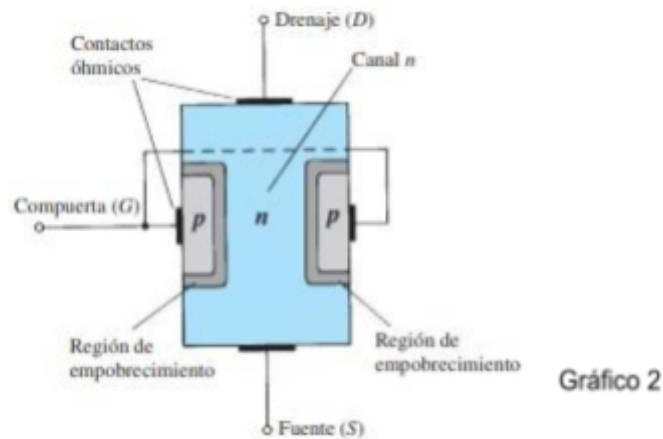
$$I_C = I_S \cdot e^{\frac{V_{BE}}{V_T}} \cdot (1 + \frac{V_{CE}}{V_A})$$

$$I_C = I_B \cdot \beta$$

$$I_E = I_C + I_B$$

The junction field-effect transistor (JFET) is a voltage-controlled device that relies directly on the presence of electrons and holes. They are generally more stable under temperature changes than BJTs and are smaller, making them ideal for integrated circuit development.

When no voltage is applied, the JFET has two depleted P-N junctions without bias, forming what is shown in Figure 2. Unlike the bipolar transistor,  $I_d$  curves can be developed as a function of  $V_{ds}$  for various gate bias levels.



*P-N junctions for NPN (a) and PNP (b) bipolar transistors.*

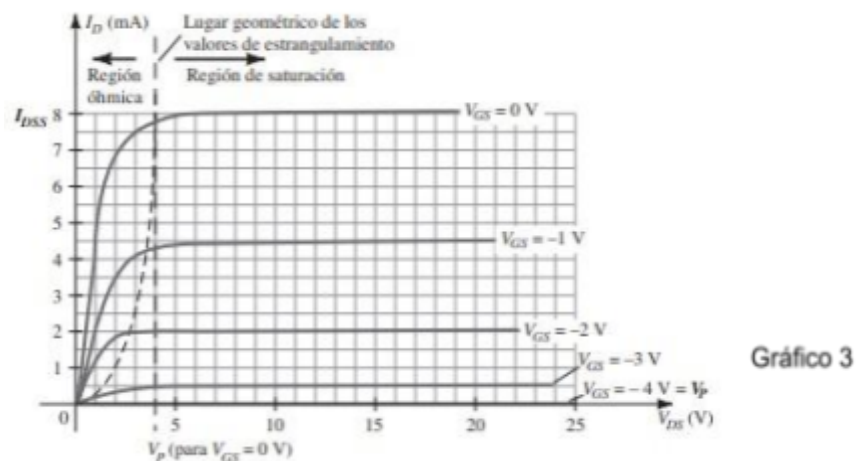
Applying a positive voltage to  $V_{ds}$  under the condition that  $V_{gs} = 0V$  generates a depletion zone near the drain to the point that the junction depletion zones "eat away" the channel. However, as  $V_{ds}$  continues to increase, a very high-density current continues to exist through an extremely small channel called  $I_{DSS}$  (obtained with  $V_{gs}=0V$ ). Thus, the drainage flow becomes:

$$I_D = I_{DSS} * (1 - \frac{V_{GS}}{V_P})^2 * (1 + \lambda * V_{DS})$$

To be able to amplify, we must always verify that the saturated channel condition is met:

$$V_{DS} > (V_{GS} - V_T) = V_{ov} > 0^1$$

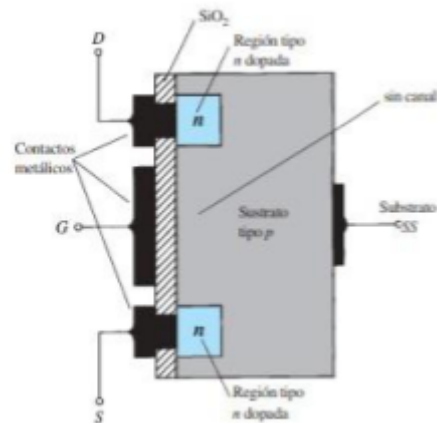
Since  $V_{gs}$  is the control voltage of the JFET, when it reaches  $V_p$ , the channel is pinched off, cut off, and no current flows. Figure 3 shows the relationship between the drain current and the  $V_{gs}$  and  $V_{ds}$  voltages.



<sup>1</sup> $V_T$  is the threshold voltage

*ID based on Vds for different Vgs values.*

On the MOSFET side (n-type), we have a structure based on a large block of p-type substrate and two small n-type regions at the drain and source terminals, as shown in Figure 4. There is an oxide layer that acts as an insulator between the metal contacts and the substrate.



*N-channel enhancement type MOSFET.*

Characterized by its high efficiency and amplification capacity, the enhancement MOSFET only allows current to flow between the drain and the source when a positive voltage is applied to the gate thanks to the generated electric field.

For  $V_{gs}$  values less than the threshold voltage, the channel ceases to exist and  $I_d$  becomes zero, being represented by:

$$I_D = \mu^* C_{ox} \frac{W}{L} * (V_{GS} - V_T)_2$$

For the field-effect transistor to amplify, it must be verified that the drain-source voltage is greater than the overdrive voltage.

## 2. TBJ

### 2.1 Specs

In the development of this first practical laboratory work, certain characteristics were taken into account when developing the single-stage designs corresponding to the two types of transistors seen in class.

For the bipolar transistor, the specifications in the PDF for group 7 were taken into account:

- 
- $|A_{VS}| = 120$
- $R_{IA} \leq 10\Omega$
- $R_S = 10\Omega$   
 $R_L = 5k\Omega$

## 2.2 Design

In this case, since it requested a high profit and a low RIA, we decided to use the common base configuration. (graph 5) Based on the BC548B, an NPN TO-92. As can be seen in Figure 6, all the components chosen to accompany the TBJ were connected on a protoboard which were later soldered to an experimental board to avoid errors.

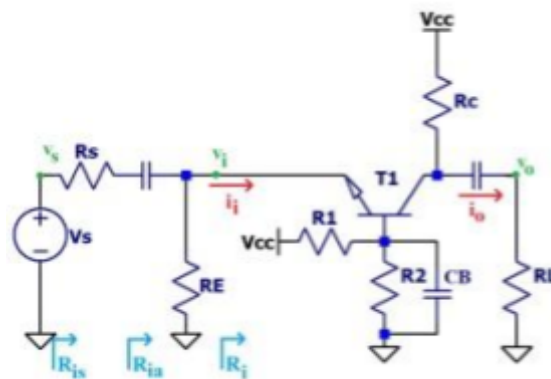


Gráfico 5

*Circuit used for the bipolar junction transistor.*

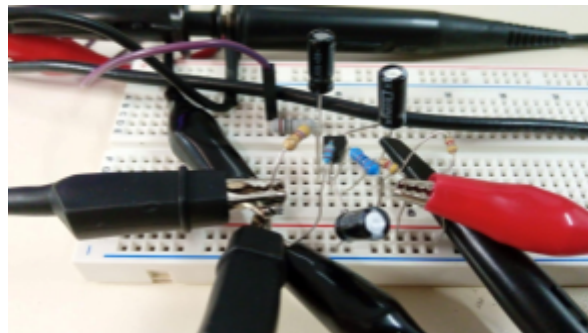


Gráfico 6

*Circuit generated in the laboratory.*

Since this topology is sensitive to variations in  $I_{CQ}$ , the value of  $I_{CQ}$  was calculated as a function of the input impedance, given that  $R_{ia}$  defines  $I_{CQ}$  in this configuration. Assuming a sufficiently high  $R_E$ , we can approximate  $R_{IA} \approx R_i = g_m^{-1}$  in conjunction with  $g_m = 40 * I_{CQ}$  to obtain an approximate collector current.

Knowing that  $R_{IA}$  must be complied with  $g_m$  the condition of  $R_{IA} \leq 10 \Omega$  we can use the approach

of  $R_{IA} \approx g_m^{-1}$  to get a minimum and not exceed the :

$$gm \leq 10 \Omega \Rightarrow gm \geq 0.1 S$$

With this condition we can calculate the minimum operating current:

$$gm \simeq 40 * I_{CQ} \Rightarrow I_{CQ} = \frac{gm}{40} = \frac{0.1}{40} = 2.5 mA$$

How 2.5mA It's the minimum current we can work with, we take  $I_{CQ} = 3 mA$

deEnenfunciónfuncióndedeesto, la current were calculated collector the parameters of the datasheet h for presented  $I^{CQ} = 3 mA$  Using in the graphic the 7 curves

- $HFE = 280$
- $HFE_{min} = 200$  •  $hie = 3150 \Omega$
- $hoe = 42 \mu S$
- $hfe = 363$

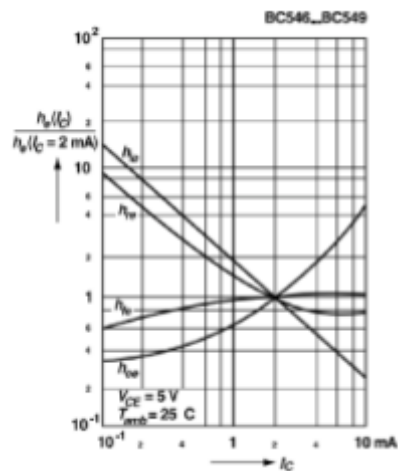


Gráfico 7

h parameters relative to the collector current.

In this way, knowing the voltage gain value, we calculate the collector resistance value.

$$|AVS| = 120 = \frac{RIA}{RIA + RS} * AV \Rightarrow AV = 120 * \frac{RIA + RS}{RIA} = 120 * \frac{20}{10} = 240$$

$$AV = gm * Rd = gm * RL || RC \Rightarrow RC = 4615.4 \Omega \Rightarrow 4k7 \Omega \text{ comercial}$$

A power supply of  $V_{CC} = 20V$  was chosen and from this, an emitter resistance was estimated in which less than 20% of  $V_{CC}$  drops.

With  $R_E$  and  $R_C$ , we can  $V_{RE}$  obtain  $\leq 20\%$  the.  $V_{CC}$  strain  $\Rightarrow$  of  $R_E$  collector =  $470 \Omega$  transmitter commercial from point Q



After trial and error, for  $V_{ceqensure} = V_{cca} - \text{independence } I_{cq} * (R_E + R_{Cof})$ , with specific 4.9 V factor of

a

shielding of approximately 100. In this way, the corresponding resistance and voltage values of the Thevenin model were obtained.

When setting up the input grid with saying 940  $\Omega$  model if  $V_{th}$  they obtained = 2.12 V the resistors of the divider

tension for the base.

$$V_{th} = V_{cc} * \frac{R_1}{R_1 + R_2} * \frac{R_{th}}{R_1} \Rightarrow R_1 = \frac{V_{cc} * V_{th}}{V_{th} - 2.12} = \frac{20V * 940\Omega}{2.12V - 2.12V} = 8867\Omega \Rightarrow R_1 = 8k2\Omega \text{ commercial}$$

Finally,  $R_{th}$  with they calculated  $R_{th1} + R_{th2} \Rightarrow R_{th2} = \frac{R_{th1} * R_{th}}{R_{th1} - R_{th}} = \frac{8.94k\Omega * 8.94k\Omega}{8.94k\Omega - 8.94k\Omega} = 1061\Omega \Rightarrow R_2 = 1k\Omega$  and  $k_{exit} = 0$  my god for the stage

amplifier.

$$R_i = h_{fe} \frac{r_{ie}}{1 + h_{fe}} = \frac{363 * 3150}{1 + 363} = 8.65\Omega$$

$$R_o = \frac{1}{h_{oe}} * (1 + \frac{h_{fe} R_E}{1 + h_{fe}}) = 421\mu S * (1 + \frac{470 * 363}{1 + 470}) = 50k59\Omega$$

Once all have been obtained the components  $R_C || R_o = 4700\Omega || 50k59\Omega$  performed = 4300  $\Omega$ . 4.7 verification of the

initial conditions based on the values of commercial resistances.

$$g_m = \frac{h_{fe}}{r_{ie}} = \frac{363}{3150} = 0.115S$$

$$R_{IA} = R_i || R_E = \frac{1}{\frac{1}{h_{fe} r_{ie}} + \frac{1}{R_E}} = \frac{1}{\frac{1}{363 * 3150} + \frac{1}{4700}} = 8.5\Omega \quad \checkmark$$

$$A_V = g_m * R_L || R_C = 0.115 * 5k + 4k = 278.6 \quad | \quad A_{VS} |$$

$$= \frac{R_{IA}}{R_{IA} + R_S} * A_V = \frac{8.5}{8.5 + 10} * 278.6 = 128 \quad \checkmark$$

$$I_{CQ} = \frac{R_{E_{th}} + V_{BE_{th}}}{R_{E_{th}} + \frac{R_{C_{th}}}{\beta_{FE_{typ}}}} = \frac{4702.017 + 0.7}{4702.017 + \frac{4700}{100}} = 3.114 \text{ mA} \quad \checkmark$$

$$V_{ceq} = V_{cc} - I_{cq} * (R_E + R_C) = 20V - 3.114 \text{ mA} * (470 + 4700) = 3.9V$$

## 2.3 Simulation

Throughout this development, the program “LTSpice” was used to perform all simulations related to both types of transistors.

Figure 8 shows the circuit made in Ltspice, along with the command used for the time-domain simulation.

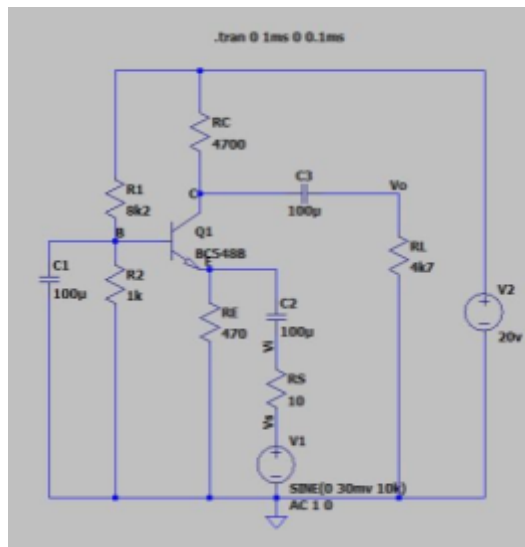


Gráfico 8

Circuit made for the TBJ BC548B

On one hand, the parameters for the DC analysis were verified, considering a frequency of 0Hz. In graph 9, it can be seen how the collector and emitter currents are extremely similar; and the current gain was calculated.

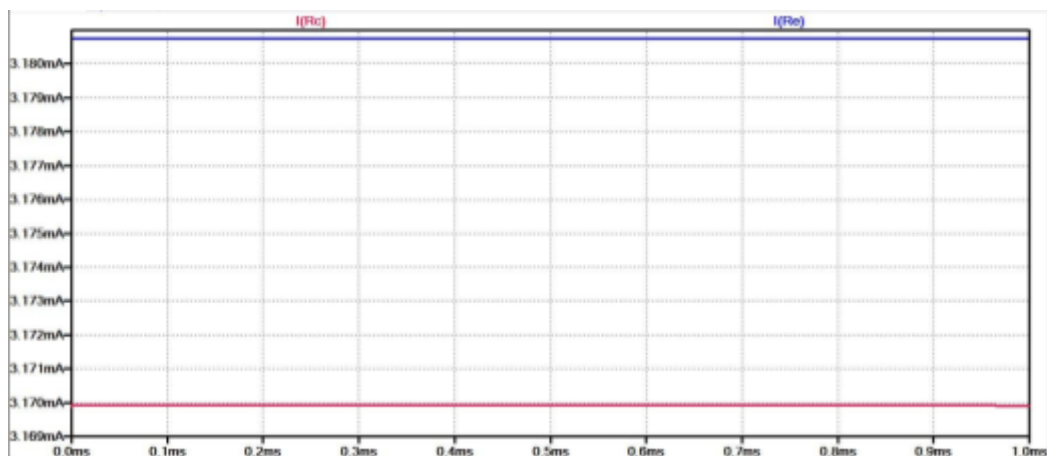


Gráfico 9

Comparison of the current passing through the collector (3,170 mA) and emitter (3,180 mA) resistors. Ai

$$=_{ie}^{ic} = \frac{3.17 \text{ mA}}{3.18 \text{ mA}} = 0.997 \text{ A/A}$$

On the other hand, the voltage drop between the collector and the emitter of the transistor was simulated, as shown in graph 10.

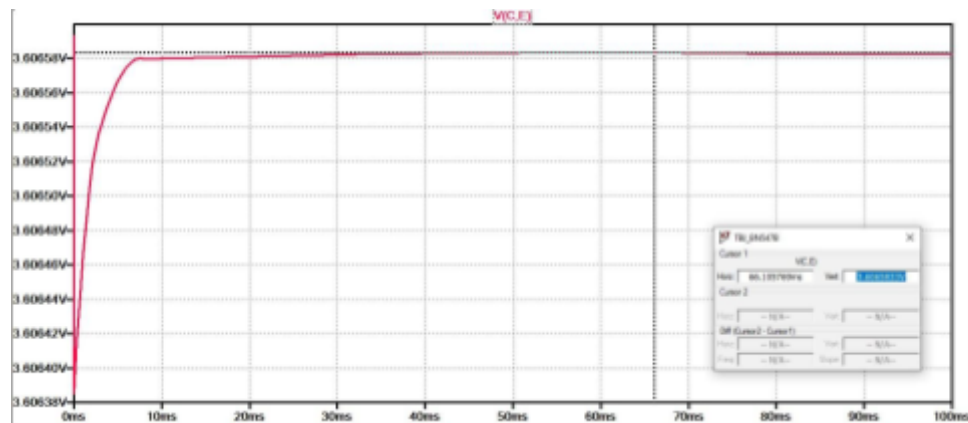


Gráfico 10

*Voltage drop between collector and emitter for the chosen design (3.6 V).*

The values of the thevenin equivalent model and the base current were also verified in graph 11.

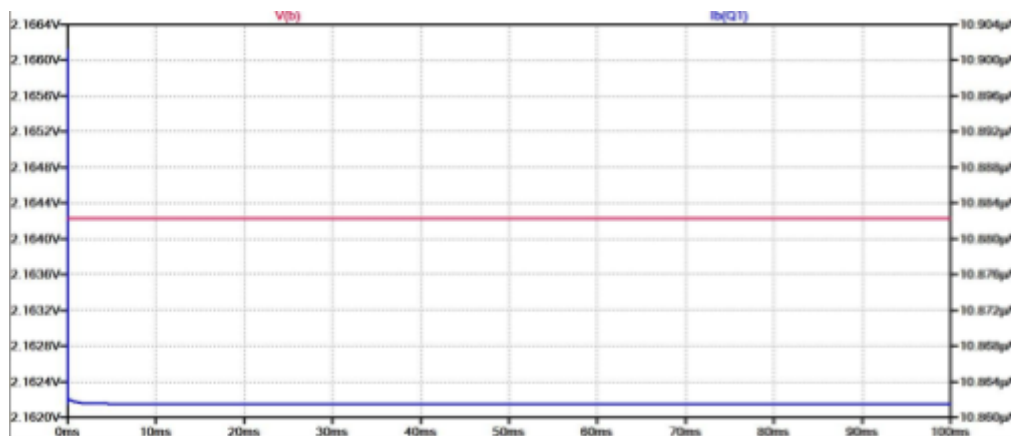


Gráfico 11

*Base current (10.8 uA) and voltage drop across resistor R2 (2.16 V)*

By studying the representation of the input and output signal, we were able to verify the voltage gain (graph 12) taking into account an input signal of 10kHz frequency.

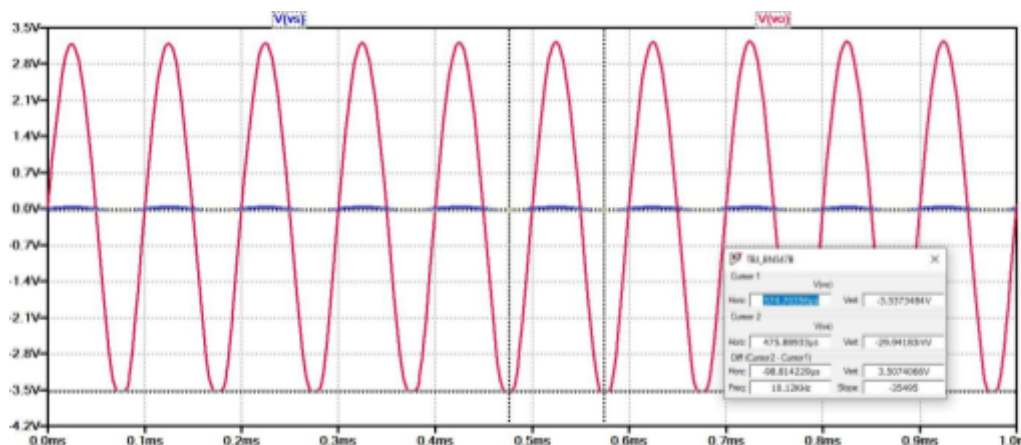


Gráfico 12

*Generator input voltage (-29.94 mV) and load output voltage (-3.53 V), with AVS=117.9V/V.*

Finally, the input resistance ( $R_{IA}$ ) and output resistance ( $R_{OA}$ ) of the amplifier stage were analyzed. In both cases, a small-signal AC simulation was used, with the difference that for the  $R_{OA}$ , the input source was passivated and a source was placed at the output, removing the load from the circuit, as shown in Figure 13.

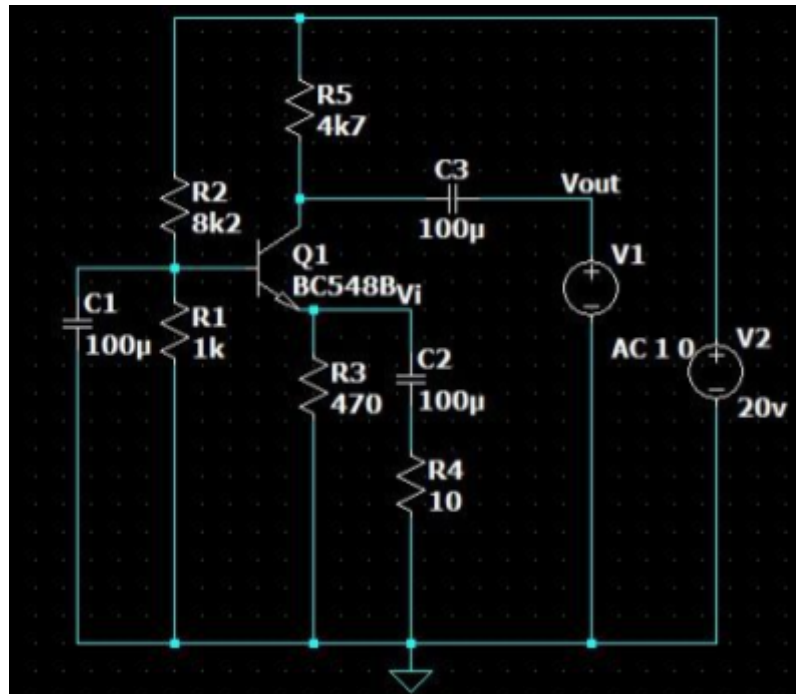


Gráfico 13

*Circuit built for ROA analysis*

The results obtained can be observed in graphs 14 and 15 and the contrast with the calculations performed.

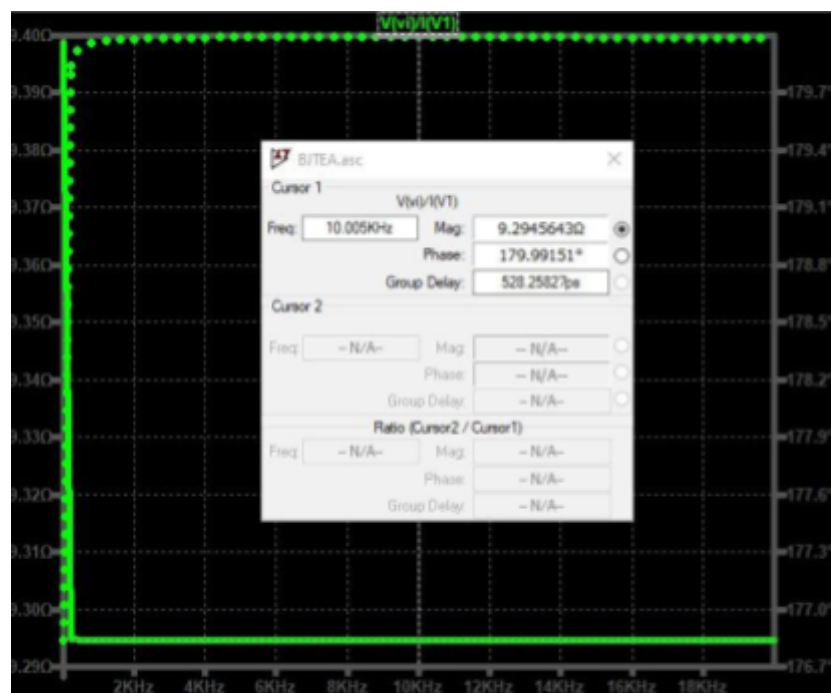


Gráfico 14

*RIA obtained in the simulation (9.290h)*

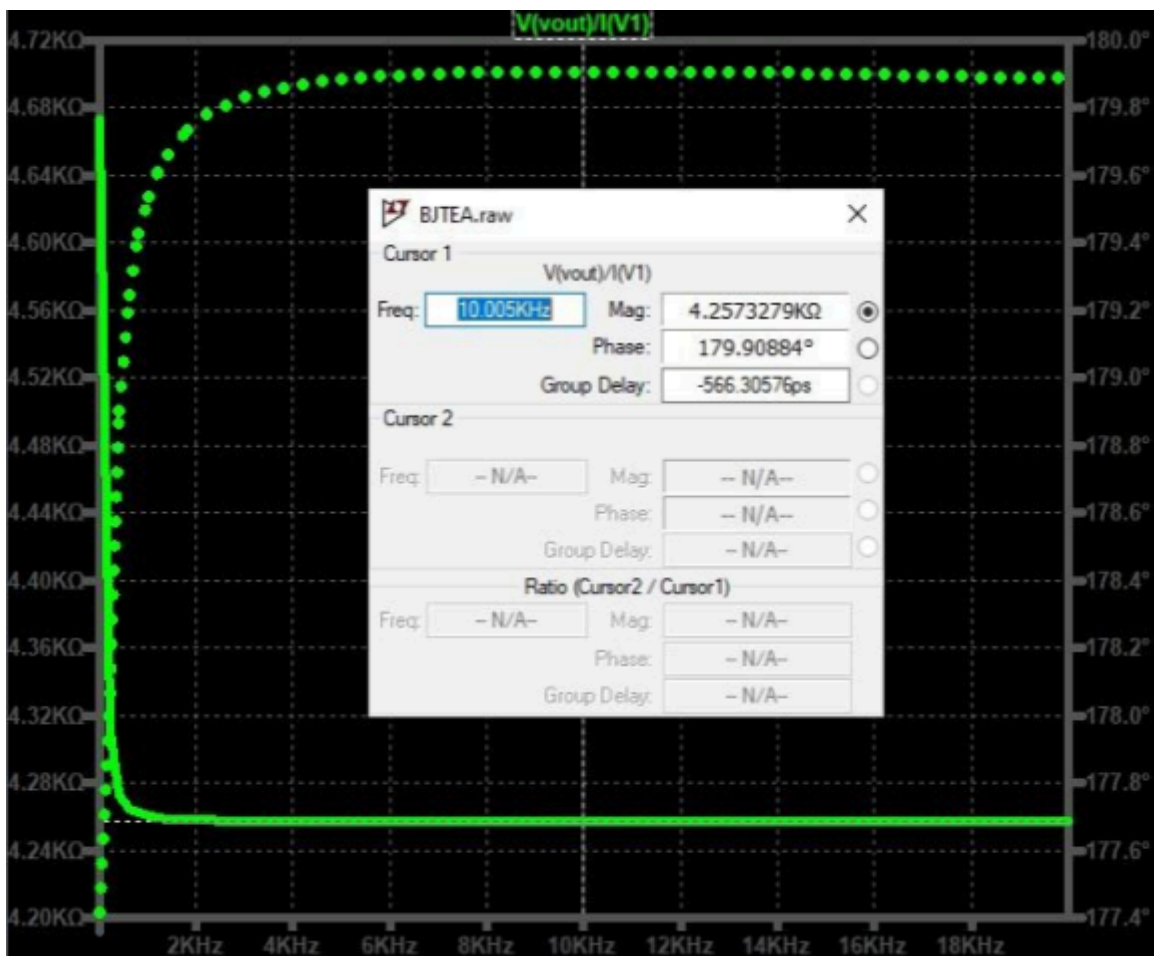


Gráfico 15

Through this simulation we were able to obtain the simulation values of the 25 Oh resistances) entrance and output to be able to compare them with the calculated ones.

$$R_{IA} = R_i || R_E \Rightarrow R_i = \frac{R_{IA} * R_E}{R_{IA} - R_E} = \frac{9.29 \text{ Oh} * 470 \text{ Oh}}{9.29 \text{ Oh} - 470 \text{ Oh}} = 9.47 \Omega$$

$$R_{OA} = R_C || R_o \Rightarrow R_o = \frac{R_{OA} * R_C}{R_{OA} - R_C} = \frac{4.25 \text{ Oh} * 4.7 \text{ Oh}}{4.25 \text{ Oh} - 4.7 \text{ Oh}} = 45164.56 \Omega$$

## 2.4 Measurements

Once the simulation has been carried out based on the values calculated on paper and pencil, all that remains is to verify the operation with real components.

A breadboard was used to position all the elements used and the resistors chosen were 1/4W and 5%.

The first thing that was checked was the voltage gain in graph 16, which was one of the main specifications imposed by the exercise.

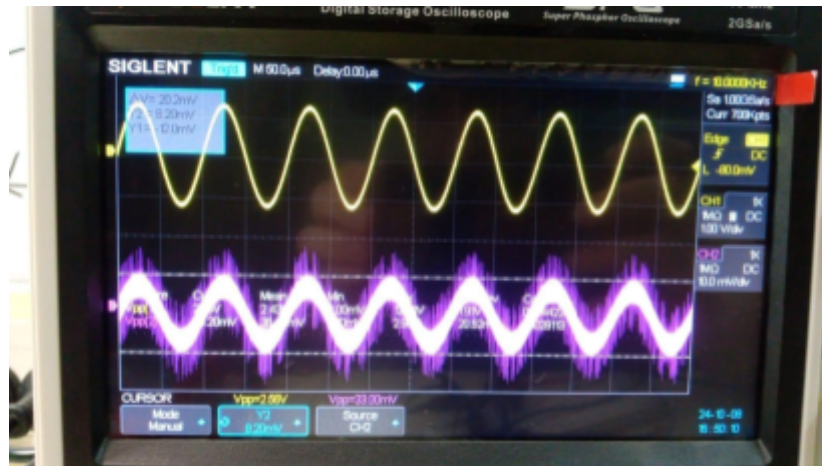


Gráfico 16

Measurement of the output signal (yellow) [2.56Vpp] and the input signal (violet), [20.2 mVpp].

$$|AVS| = \frac{V_o}{V_s} = \frac{2560}{20.2} = 127.72$$

Since a fairly low amplitude was chosen for the signal generator, the presence of noise noticeably affected the input signal, and cursors were used to obtain a more precise value.

The corresponding numbers for the static working point and some intermediate values used in the previous calculations were also verified.

Using a multimeter, the voltage drop across the collector resistor (graph 17) and the emitter resistor (graph 18) was measured, and the operating current was verified.



Gráfico 17



Gráfico 18

Measurement of voltage drops across the collector resistor (14.84 V) and across the emitter resistor (1.508 V).

$$V_{RC} = 14.84 \text{ V} \Rightarrow I_{Cq} = \frac{V_{RC}}{R_C} = \frac{14.84}{4700} \text{ A} = 3.16 \text{ mA}$$

$$V_{RE} = 1.508 \text{ V} \Rightarrow I_{Cq} = \frac{V_{RE}}{R_E} = \frac{1.508}{470} \text{ A} = 3.2 \text{ mA}$$

$$I_{Cq} = (3.16 \text{ mA} + 3.2 \text{ mA}) = 3.18 \text{ mA}$$

Thanks to these empirical values,  $g_m \approx 40 \text{ S}$  capable\*  $I_{CQ} = 0.1272 \text{ A}$  obtain  $S$  the voltage drop between the

collector and emitter.

$$V_{CC} = V_{RC} + V_{CE} + V_{RE} \Rightarrow V_{CE} = V_{CC} - V_{RC} - V_{RE} = 20V - 15.18V - 1.54V = 3.28V$$

Likewise, the voltage drop across the bias resistor R2 was measured to verify the Thevenin equivalent, as can be seen in Figure 19.



Gráfico 19

Measurement of the voltage drop across R2 (2.179 V).

Once these values were obtained, the current relationship could be verified, taking into account that the current gain for this configuration theoretically tends to one.

After all the calculations and measurements  $A_i = i_{eic} = 33.2723 \text{ mA/A}$  until 0.987 is  $A/A$  point, we study the limits

cutoff and saturation as well as the maximum excursion of  $V_o$ . From defining the following limits given as:

Cut-off limit  $\rightarrow b = I_{CQ} \cdot R_{din}$  where  $R_{din} = R_c // R_L$

Saturation limit  $\rightarrow a = V_{ceq} - V_{cesat}$  where  $V_{cesat} = 0.25V$  saturation  $I_{CQ}$  is left

$$a = 3mA \cdot 4.25V = 12.75mV \quad b = 7.268V$$

Where  $a$  and  $b$ , we see that by cutting it remains  $a$  and by



We observed the empirical values of  $V_{ceq}$  and  $I_{cq}$  and we see that because  $V_{ceq} = 3.03V$  and  $I_{cq} = 3.18mA$  please observe carefully  $V_{ceq} = 3.03V$  and  $I_{cq} = 3.18mA$

With  $V_{ceq} = 3.03V$  almost difference at the limit due to the cutoff, as it runs about  $0.45V$  while a change is noticeable in the saturation limit, which shifts by almost  $1.2V$ . Therefore, the maximum symmetrical excursion  $V_{omax} = 4.25V$  with calculated data and given that in both cases it turns out that the transistor limits by saturation  $V_{omax} = 3.03V$  with empirical data.

We now observe the graph of the static (RCE) and dynamic (RCD) load lines in graph 20.

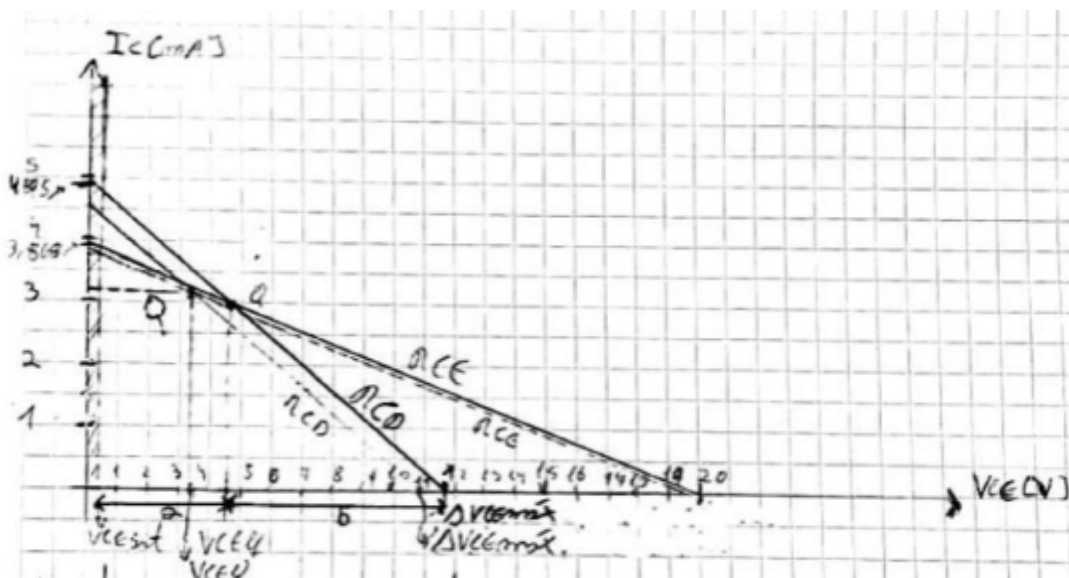


Gráfico 20

RCE and RCD for calculated data in solid line and for empirical data in dotted line.

### Calculated data:

Si  $I_{cq} = 0A \Rightarrow V_{ceq} = V_{cc} = 20V$ , y si  $V_{ceq} = 0V \Rightarrow I_{cq} = \frac{V_{cc}}{R_E + R_C} = 3.8685mA$

Por otro lado, el punto Q queda  $Q = \{V_{ceq}, I_{cq}\}$  donde  $V_{ceq} = 4.5V$  y  $I_{cq} = 3mA$  Además

$$\Delta I_{cm\acute{a}x} = \frac{\Delta V_{cem}}{R_{din}} = \frac{V_{cem}}{R_C // R_L} = \frac{V_{ceq}}{R_C // R_L} + b = \frac{4.5}{2422} + 1.86 \cdot 10^{-6} = 4.895mA$$

### Empirical data:

Si  $I_{cq} = 0A \Rightarrow V_{ceq} = V_{cc} = 20V$ , y si  $V_{ceq} = 0V \Rightarrow I_{cq} = \frac{V_{cc}}{R_E + R_C} = 3.8685mA$  Por otro lado, el

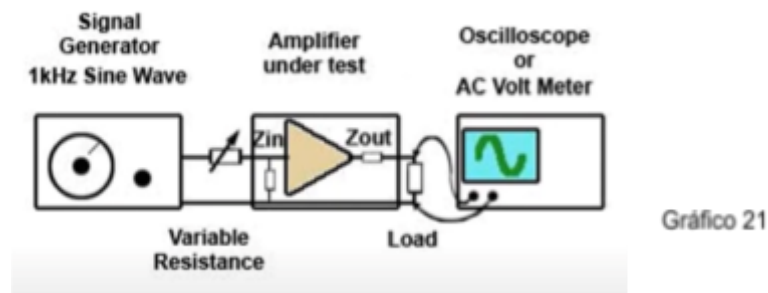
punto Q queda  $Q = \{V_{ceq}, I_{cq}\}$  donde  $V_{ceq} = 3.28V$  y  $I_{cq} = 3.18mA$



Besides,  $\Delta I_{cm}$  depends of the generator  $\Delta V_{cem} R_{din} \approx \Delta R C_{signs} V_{cem} // R_L x$  and a  $V_{ceq} R C_{oscilloscope} // R_L + b = 242210$ . went  $96.68 V \Omega =$  capable  $4.532 mA$  of measure the

input and output impedance, using the methods suggested in class.

Regarding the input impedance of the stage, we refer to Figure 21. Using a 1kHz signal generator, we aim to maximize the oscilloscope output. The key lies in the setup and a potentiometer; varying its resistance alters the waveform of the output signal at the load. We repeat this process until the output signal reaches half the amplitude of the original signal measured without the potentiometer.



*Circuit used to measure the input impedance of the design*

Finally, the various measurements that we can observe in the following graphs 22, 23, 24 are obtained.



*Vout in RL with function generator at the stage input without putting potentiometer at the input (1.4 V).*



Gráfico 23

$V_{out}$  in  $R_L$ , with function generator at the stage input and with potentiometer at the input. (704 mV).

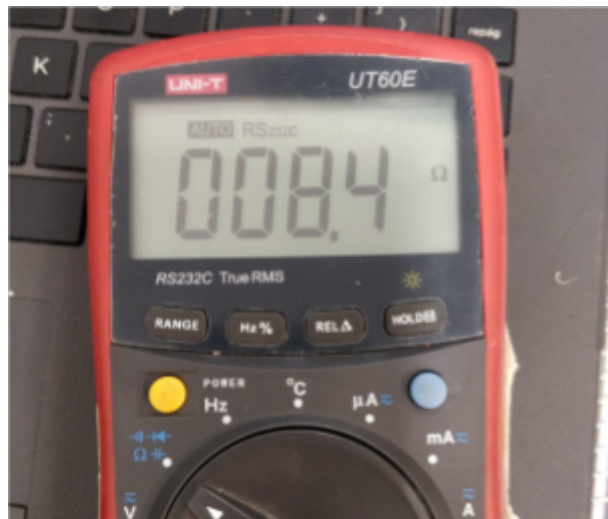


Gráfico 24

Input impedance  $Z_i$  measured with the potentiometer disconnected.

We see that  $Z_i = 8.4 \text{ Ohms}$  approximately, which is close to the input resistance of the stage calculated above.

Regard  $R_{IA}$  (a calculated impedance) =  $8.05 \Omega$  exit  $R_{IA}$  the process (simulated) varies = and  $9.1 \text{ bit}$ ,  $29 \Omega$  measuring  $R_{IA}$  (measured) the change =  $8 \text{ in } 4 \Omega$  the

output when connecting the load resistor, as shown in graph 25. To do this, we measure the voltage in “open circuit” (A) and the voltage in “circuit with load” (B).

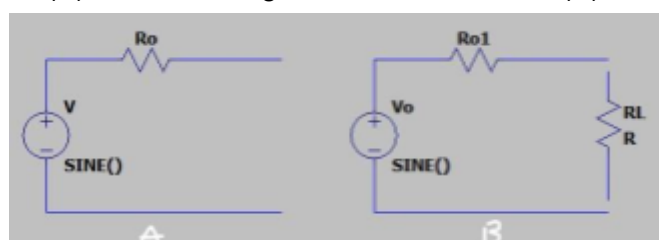


Gráfico 25

Open circuit (A) and circuit with load (B).

The measurements mentioned in the following graphs 26, 27 are obtained.



Gráfico 26

$V_{out}$  without  $R_L$  with function generator at the input of the stage in open circuit. (99.6 mV).



Gráfico 27

$V_{out}$  in  $R_L$  with function generator at the input of the stage in loaded circuit. (49.6 mV).

Finally, to calculate the output resistance of the stage, we simply substitute the values into the equation that relates the voltages:

$$V_{open} = 99.6 \text{ mV}$$

$$V_L = 49.6 \text{ mV}$$

$$Z_o = R_L * \left( \frac{V_{open}}{V_L} - 1 \right) = 4k7\Omega * \left( \frac{99.6 \text{ mV}}{49.6 \text{ mV}} - 1 \right) = 4737.9\Omega$$

Also  $ROA$  (calculated) is possible) = 4300 repeat. 47he0hprocess  $ROA$  of the (simulated) endurance) = of 4k Prohibited, 25  $\Omega$   $ROA$  varying (measured) the value) = 4737 of the. 9  $\Omega$

load resistance until the voltage of the loaded circuit is half the voltage of the open circuit.

$$V_L = \underline{v_{open2}}$$

## 3. JFET

### 3.1 Specs

For the field-effect transistor, it was specified that:

- $A_{VS} > 7$
- $R_{IA} \leq 450\Omega$
- $R_S = 330\Omega$   
 $R_L = 15k\Omega$

## 3.2 Design

In the case of the field-effect transistor, a low gain and a low output impedance were needed, so it was decided to use a common gate and assemble the circuit shown in graphs 28 and 29.

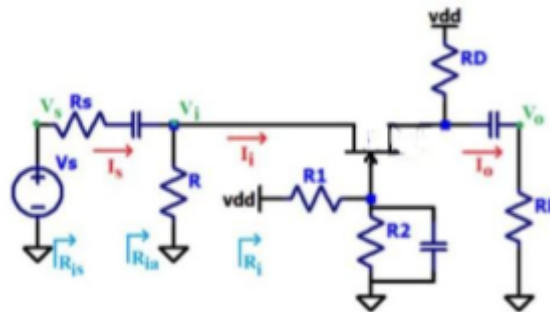


Gráfico 28

Circuit chosen for the field-effect transistor

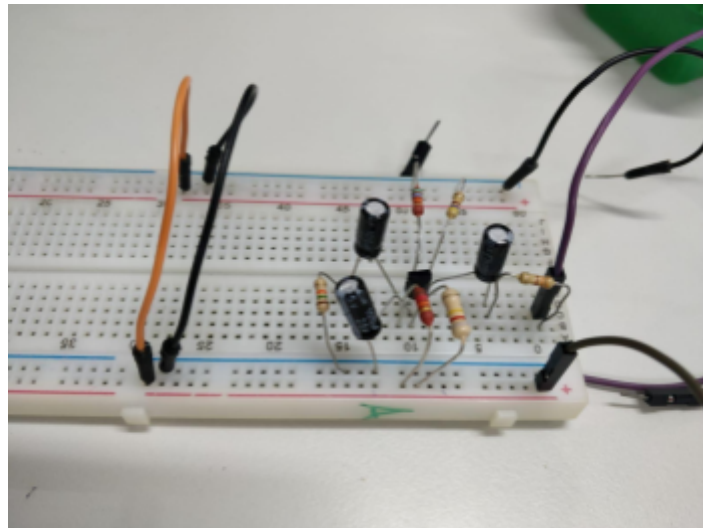


Gráfico 29

Circuit assembly for the JFET.

The  $I_{DSS}$  value can be obtained more precisely from the data sheet by looking at the  $I_{ds} = f(V_{ds})$  graph. From there, find the curve with  $v_{gs} = 0$ , which is usually the one with an almost completely horizontal slope. The current value for that curve is  $I_{DSS}$ . However, if no such graph is available, consult the "on characteristics" section to see the acceptable range of  $I_{DSS}$ .

### ON CHARACTERISTICS

|                                                                                                  |           |     |    |      |
|--------------------------------------------------------------------------------------------------|-----------|-----|----|------|
| Zero – Gate – Voltage Drain Current(1)<br>( $V_{DS} = 15\text{ Vdc}$ , $V_{GS} = 0\text{ Vdc}$ ) | $I_{DSS}$ | 2.0 | 20 | mAdc |
|--------------------------------------------------------------------------------------------------|-----------|-----|----|------|

On the other hand, to obtain  $V_p$  = Pinch Off Voltage, we were able to obtain it from the "off characteristics" section. There,  $V_{GS(off)}$  is mentioned, and this value is equal to  $V_p$ .

The value suggested in class for this range was 3.5V, which is an intermediate value in this range.

|                                                               |          |   |       |     |
|---------------------------------------------------------------|----------|---|-------|-----|
| Gate – Source Cutoff Voltage<br>(VDS = 15 Vdc, ID = 2.0 nAdc) | VGS(off) | — | – 8.0 | Vdc |
|---------------------------------------------------------------|----------|---|-------|-----|

The data sheet used as a reference was that of “On Semiconductors” as it is the most complete compared to other brands such as Motorola, Fairchild, etc.

We can ensure, through experience, that we can use the current dispersion of calculations, and that the work took an approximate error of our own.  $ID_{SS} = 16 \text{ these } 4 \text{ mA device } ID_{SS} V_{p2} = -7.35 \text{ no. } 5 \text{ mA } V$  They affect us.

$$I_D =$$

Knowing the  $ID_{SS}$  and  $V_p$ , we obtained the voltage drop between the gate and the source, as well as the value of  $gm$ .

$$I_D = ID_{SS} * (1 - \frac{V_{gs}}{V_p})^2 \Rightarrow 7.5 \text{ mA} = 16.4 \text{ mA} * (1 - \frac{V_{gs}}{3.5V})^2 \Rightarrow V_{gs} \simeq -1.38 \text{ V}$$

Having  $gm = 2 * \frac{ID_{SS}}{V_{p2}}$  the value ( $V_{gs}$  of the  $V_{p2}$  endurance) =  $2 * \frac{16.4 \text{ mA}}{3.5 \text{ V}}$  Prohibited)  $2 * (-1.38 \text{ V} - 3.5 \text{ V})$  the condition =  $6.33 \text{ mS}$  to

By measuring voltage gain, we were able to obtain both the  $R_{so}$  and  $R_D$  values. Since the

$R_{IA}$  had to be less than 450, one of 150 was chosen.  $R_i \simeq \frac{1}{gm} = \frac{1}{6.33 \text{ mS}} = 158 \Omega$

$$R_{IA} = R_i || R = 150 \Rightarrow R = 2962.5 \Omega \Rightarrow 2k7 \Omega \quad 10 < AVS = R_{IA}^{R_{IA} + R_S} * (gm * R_D) =$$

$$\frac{R_{IA}^{R_{IA} + R_S} * gm * R_L || R_D}{R_{IA}^{R_{IA} + R_S} * gm * R_L || R_D} \Rightarrow R_D > 7k6 \Omega \quad 5K5$$

We define a power supply of  $R_D 20V$ , = to  $10k\Omega$  then obtain the value of  $R_{so}$  by searching that its voltage drop is less than 20% of  $V_{DD}$ .

$$V_{R_{so}} < 0.2 * 20 = 4 \text{ V}$$

We also obtain the tension  $R < 4I_{\text{Drain-Source}} V = 7.54 \text{ mA} V$  = of the 533 point.  $3\Omega Q. \Rightarrow R = 470 \Omega$

$$V_{DSQ} = V_{DD} - I_D * (R + R_D) = 20V - 7.5mA * (470\Omega + 10k\Omega) = -58.5V$$

It can be observed that with such a high current, the voltage drop across the resistors is very high and the power supply would go out of range, which is why we initially chose a current close to 1mA despite going against what was seen in class (very close to the lower limit) with the parameters seen in the PDFs.  $R_{IA} = R_i || R \simeq g_m^{-1} \Rightarrow g_m = R_{IA}^{-1} = 2.22mS$

$$g_m = 2 * \frac{I_{DSS}}{|V_{p}|} * \sqrt{\frac{I_D}{I_{DSS}}} \Rightarrow I_D = (g_m * \frac{|V_{p}|}{2})^2 * I_{DSS} = (2.22mS * \frac{2V}{2})^2 * 12mA = 924\mu A$$

$$I_D = I_{DSS} * (1 - \frac{V_{gs}}{V_p})^2 \Rightarrow V_{gs} \simeq -2.17V$$

$$R_i \simeq \frac{1}{g_m} = \frac{1}{2.22mS} = 450.45\Omega$$

$$8 < AVS = \frac{R_{IA}}{R_{IA} + R_S} * (g_m * R_D) = \frac{R_{IA}}{R_{IA} + R_S} * g_m * R_L || R_D = \frac{450}{450 + 330} * 2.22mA * 15k || R_D \Rightarrow R_D > 10k7$$

Continuing with a power supply  $R_D = 2012 \ln k: 0h$

$$V_{R_{SO}} < 0 * 20 = 4V$$

We also obtain the tension  $R < 4I_{\text{Drain-Source}} V = 924 \mu A$  of the 4330 point  $0h \Rightarrow Q.R = 3900\Omega$

$$V_{DSQ} = V_{DD} - I_D * (R + R_D) = 20V - 924\mu A * (3900\Omega + 12k\Omega) = 5.30V$$

Finally, to decide the voltage divider for the biasing, a large value of R1 was imposed and the Thevenin model was performed to obtain an equivalent based on the input mesh.

$$R1 = 820k\Omega$$

$$V_{th} = V_{gs} + I_D * R = (-2.17V) + 924 \mu A * 3900\Omega = 1.43V$$

$$V_{th} = V_{DD} * \frac{R_2}{R_2 + R1} \Rightarrow R2 = \frac{V_{th} * R1}{V_{CC} - V_{th}} = \frac{1.43V * 820k}{20V - 1.43V} = 63k14\Omega \Rightarrow R2 = 69k\Omega$$

In addition, the input and output impedance values for the common gate were calculated.

$$r_o = \frac{1}{g_{os}} = \frac{1}{0.035 \text{ m}} \simeq 28571.42 \Omega$$

$$R_d = R_D || R_L = 12 \text{ k}\Omega || 15 \text{ k}\Omega \simeq 6 \text{ k}66 \Omega$$

$$R_o = r_o * (g_m * (R || R_S) + 1) + R || R_S =$$

$$28571.42 \Omega * (2.22 \text{ mS} * (3 \text{ k}9 \Omega || 330) + 1) + 3 \text{ k}9 || 330 = 33147.6 \Omega$$

Now that we have the calculated values,  $ROA = \text{with } R_o || \text{performed } R_D = \text{the } 33147 \text{ verification.}$   
 $6 \Omega || 15 \text{ corresponding } k\Omega = 10 \text{ k}3 \Omega$  of the

Specifications. It is clarified that the obtained “ro” has a certain error since gos was obtained from the datasheet by observing the associated logarithmic graph, obtaining the data approximately.

$$V_{th} = V_{DD} * \frac{R_2}{R_2 + R_1} = 20 \text{ V} * \frac{69 \text{ k}\Omega}{69 \text{ k}\Omega + 820 \text{ k}\Omega} = 1.55 \text{ V}$$

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{69 \text{ k}\Omega * 820 \text{ k}\Omega}{69 \text{ k}\Omega + 820 \text{ k}\Omega} = 63 \text{ k}64 \Omega$$

$$R_i = \frac{1}{\frac{1}{r_o g_m} + \frac{1}{R_d}} = \frac{1}{\frac{1}{28571.42 \text{ mA} * 28571} + \frac{1}{6 \text{ k}66.42}} = 547 \Omega$$

$$V_{th} = I_D * R + V_{gs} \Rightarrow I_D = \frac{V_{th} - V_{gs}}{R} = \frac{1.55 \text{ V} - 0 \text{ V}}{3900 \Omega} = 953.8 \text{ uA} \checkmark$$

$$R_{IA} = R_i || R = 547 \Omega || 3900 \Omega = 479.7 \Omega \checkmark$$

$$AV = \frac{R_d}{r_o + R_d} * \frac{R_o}{R_o + R_S} = \frac{6 \text{ k}66 \Omega}{28571.42 \text{ mS} * 6 \text{ k}66 \Omega + 1} = 12.18$$

As was proven in |paper AVS| = and in  $R_{IA}$  calculations +  $R_S * AV_{no} = \text{we}$   
 achieved  $479.8 \Omega + 8330 \Omega \text{ complete } 0 \text{ h} * 12.18 \text{ all} = 7.21 \text{ the } \checkmark \text{ specs}$

Within a useful working range, precisely for this reason, and weeks of attempts with different values and configurations, we decided to perform the analysis for a common source according to graph 30.



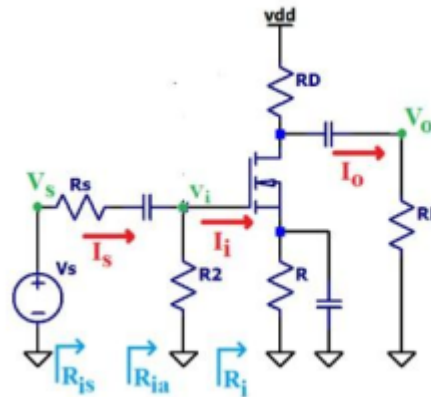


Gráfico 30

Configuration chosen in a second instance for the JFET

In the first case, we tried with the typical values seen in class for JFETs ( $ID_{SS} = 12 \text{ mA}$  and  $V_p = -3 \text{ V}$ ).

$$I_D = ID_{SS} * (1 - \frac{V_{gs}}{V_p})^2 \Rightarrow 6 \text{ mA} = 12 \text{ mA} * (1 - \frac{V_{gs}}{-3 \text{ V}})^2 \Rightarrow V_{gs} \simeq -0.88 \text{ V}$$

Having  $g_m$  the value = 2 of  $ID_{SS}$  the  $p_z$  endurance\* ( $V_{gs} - V_{pof}$ ) load, = 2 \* the  $(-R_{IA} 153. \text{mA} 5 \text{ V})$  and  $z$  and\* (approximate - 1.1 V -- of 3.5 the) = revenue 5.65 mS what with

We search, we obtain the rest of the resistances and the Vds.

$$R_{IA} = 450 \Omega \Rightarrow R_2 = 470 \Omega$$

$$R_o = r_o = \frac{1}{g_m} = \frac{1}{2 \text{ mS} * 6 \text{ mA}} \simeq 83.3 \text{ k}\Omega$$

$$A_{VS} = \frac{R_{IA}}{R_{IA} + R_S} * (-g_m * R_L || R_D || R_o) \Rightarrow R_D = 3089 \Omega \Rightarrow R_D = 2700 \Omega$$

$$R_{so} = 150 \Omega \text{ (provided it is less than 20\% of VDD).}$$

$$R_{OA} = r_o || R_D = 83.3 \text{ k}\Omega || 2.7 \text{ k}\Omega \simeq 2615 \Omega$$

We checked  $V_{ds} = V_{DD} - I_D * (R_{so} + R_D) = 20 \text{ V} - 6 \text{ mA} * (150 \Omega + 2700 \Omega) = 2.9 \text{ V}$

$$-V_{gs} = R_{so} * I_D \Rightarrow I_D = \frac{-V_{gs}}{R_{so}} = \frac{-0.88 \text{ V}}{150 \Omega} = 5.87 \text{ mA} \quad \checkmark$$

$$R_{IN} = R_2 = 470\Omega \quad \checkmark$$

$$|A_{VS}| = \frac{R_{IN}}{R_{IN} + R_S} \cdot (g_m \cdot R_L || R_D || R_o) \Rightarrow 7.4 \quad \checkmark$$

To obtain the expected results in practice, we decided to recalculate the components for a completely new transistor. Using the 2N3958 JFET, an N-channel transistor specifically designed for amplification applications, the following analysis was performed:

With  $V_{GS} = -2V$  and  $I_{DSS} = 3mA$  and following what was explained in class for the current of

$$I_D = I_{DSS} \cdot \left(1 - \frac{V_{GS}}{V_P}\right)^2 \Rightarrow 1.5 mA = 3 mA \cdot \left(1 - \frac{V_{GS}}{-2V}\right)^2 \Rightarrow V_{GS} \simeq -0.59V$$

From the data sheet,  $V_P = -2V$  and  $I_{DSS} = 3mA$ . We calculate

the remaining resistances and the voltage drop between the drain and the source.

$$R_{IA} = 450\Omega \Rightarrow R^2 = 470\Omega$$

$$R_o = r_o = \frac{1}{g_o} = \frac{1}{4.25mS} \simeq 238\Omega$$

$$A_{VS} = \frac{R_{IN}}{R_{IN} + R_S} \cdot (-g_m \cdot R_L || R_D || R_o) \Rightarrow 7.4 < \frac{470\Omega}{470\Omega + 330\Omega} \cdot (-2.12mS \cdot 15k\Omega || R_D || 78k4\Omega)$$

$$R_D \simeq 11k5\Omega \Rightarrow \text{what? } 10k\Omega \text{ smaller? } 12k\Omega$$

$$R_{SO} = 390\Omega \quad (20\% \text{ of } V_{DD}).$$

$$L = R || N = 78k4 || 10k\Omega \simeq 8k8\Omega$$

$$V_{DS} = V_{DD} - I_D \cdot (R_{SO} + R_D) = 20V - 1.5mA \cdot (390\Omega + 10k\Omega) = 4.415V$$

We verified the calculation.

<sup>2</sup>Using a potentiometer







✗



Drain and source current for the selected components (926.3uA).

The voltage drop between drain and source was also verified in graph 33 and the equivalent value for the thevenin model of the input in graph 34.

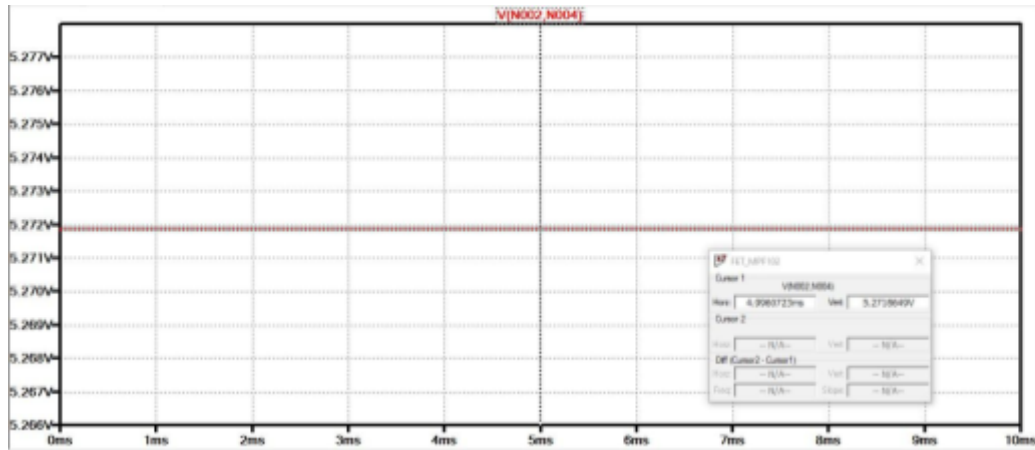


Gráfico 33

Voltage between drain and source (5.27V).

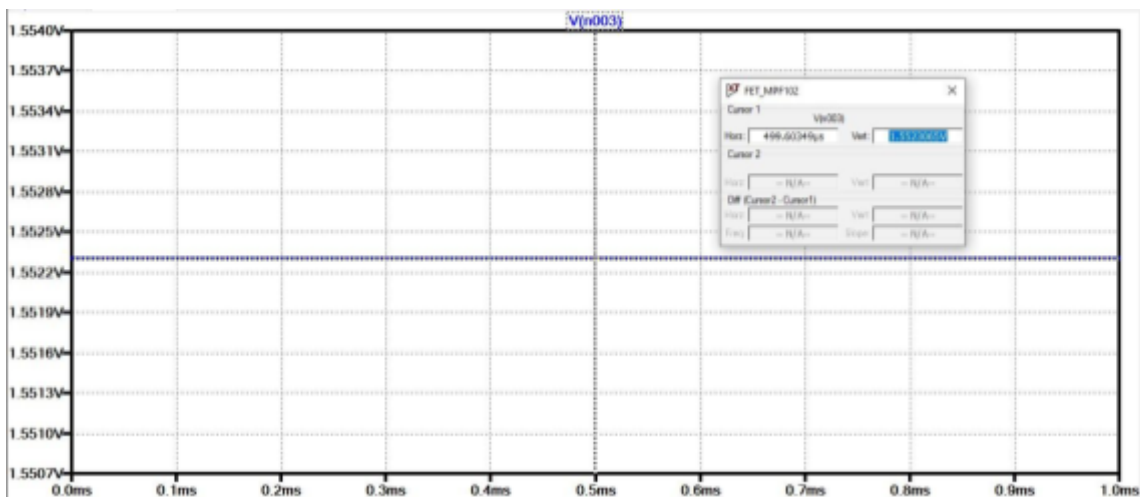


Gráfico 34

Voltage drop across resistor R2 (1.55V).

The voltage drop between the gate and the source (graph 35), the maximum drain current (graph 36) and the Vp for this circuit with JFET were verified.

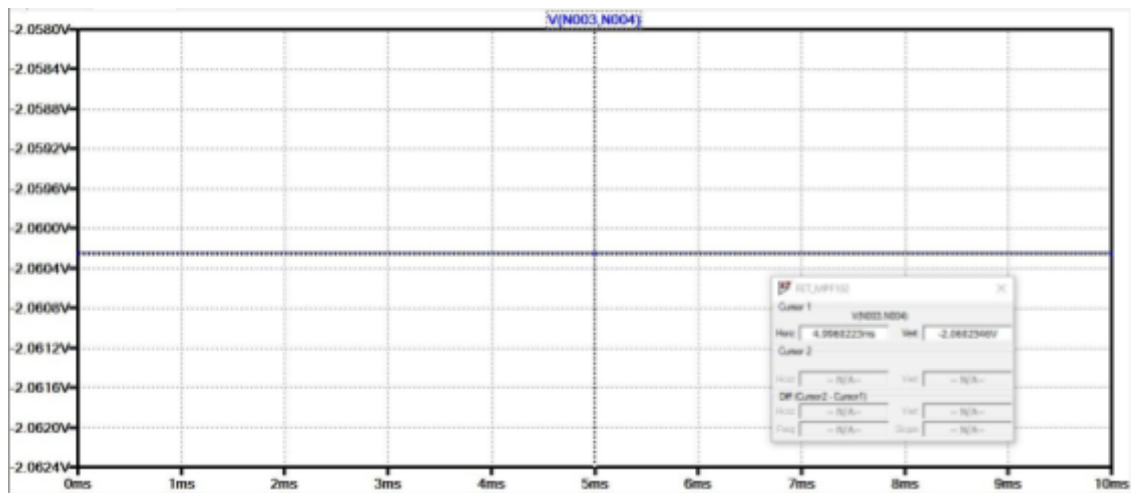


Gráfico 35

Voltage drop between the gate and the source (-2.1 V).

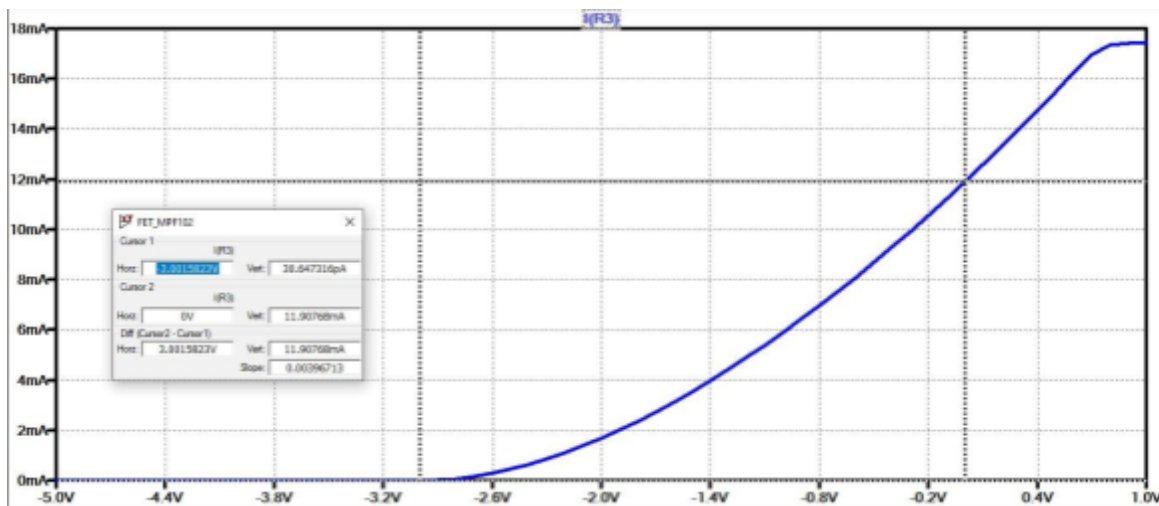


Gráfico 36

Construction characteristics,  $I_{dss}$  (11.91 mA) and  $V_p$  (-3 V).

In this way, we were able to verify the value of  $g_m$  in simulation.

$$g_m = 2 * \frac{I_{dss}}{V_p} * (V_{gs} - V_p) = 2 * \frac{11.91 \text{ mA}}{-3 \text{ V}} * (-2.17 \text{ V} - (-3 \text{ V})) = 2.19 \text{ mS}$$

Last but not least, we were able to simulate the voltage gain across the input and output signals of the circuit, as can be seen in Figure 37.

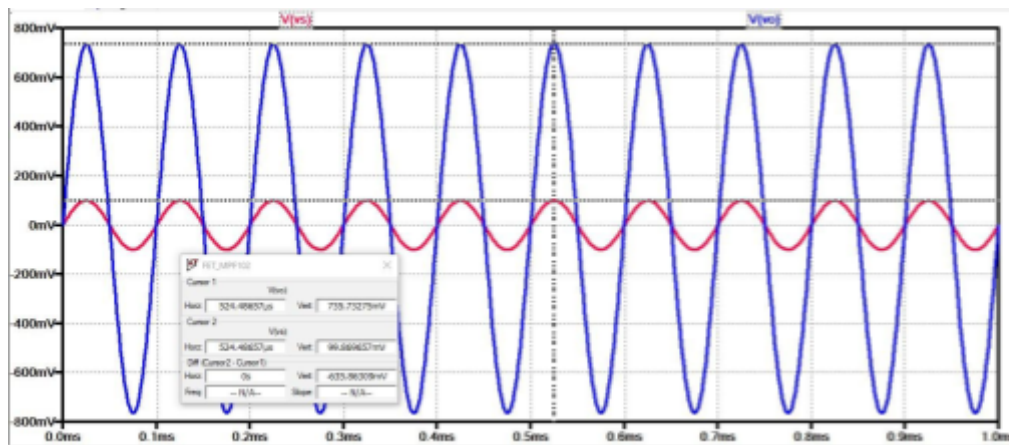


Gráfico 37

Generator input voltage (-99.87 mV) and load output voltage (-735.7 mV), with  $AVS=7.37$  V/V.

The ROA and RIA measurements are taken in exactly the same way as with the TBJ. The results obtained for RIA are presented in Figure 38 and for ROA in Figure 39.

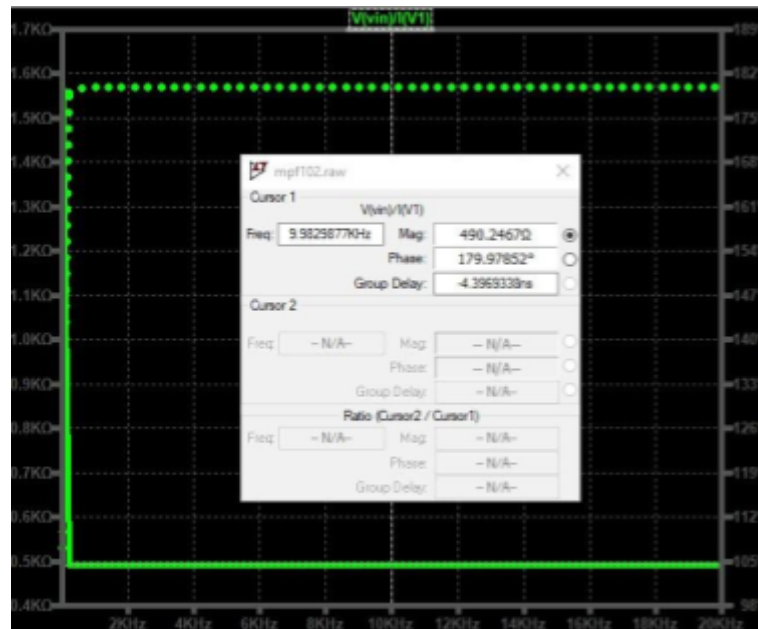


Gráfico 38

RIA obtained in the simulation (490.240h)

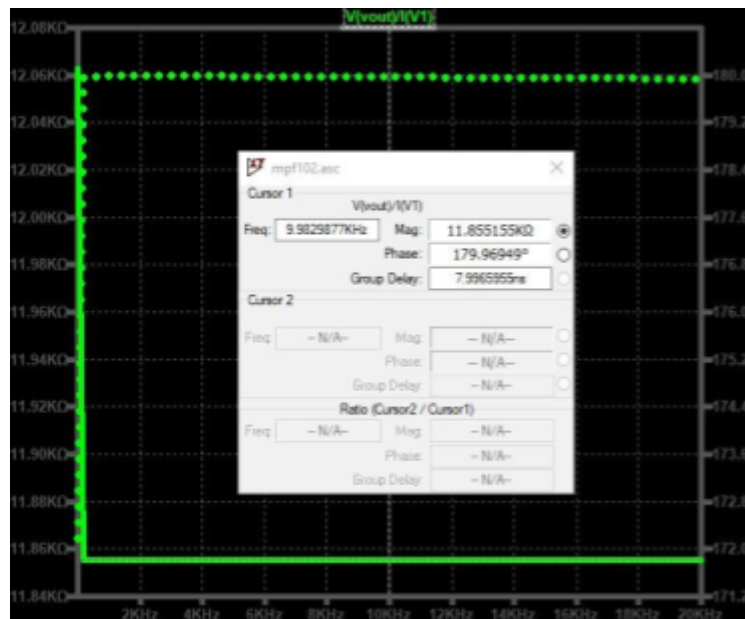


Gráfico 39

In the case of resistance of LONG Prohibited obtained with in observe the simulation what (11k85 fulfills Oh with the calculation

Previously, while for ROA the value varies but is in the same order of magnitude.

As before, we obtained the input and output resistance values so we could compare them with the calculated ones.

$$R_{IA} = R_i || R \Rightarrow R_i = \frac{R_{IA} * R_{So}}{R_{IA} - R_{So}} = \frac{490 * 240\Omega}{490 - 5600} = 537.27\Omega$$

$R_o = r_o * (g_m * (R_{So} || R_S) + 1) = 28571.42\Omega * (1.7mS * (5600\Omega || 330) + 1) = 43708\Omega$  We also performed the corresponding simulations for the common source configuration model according to graph 40.

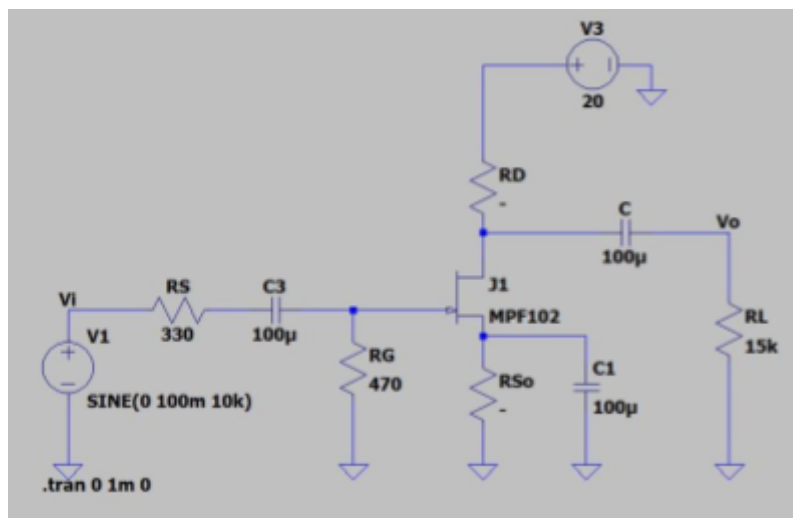


Gráfico 40

Circuit made for the MPF102 JFET.

We decided to use a 10k drain resistor for the simulation due to the similarity of the results to what we expected.

In graph 41 we present the gain obtained in the simulation and in graph 42 the working current, the voltage drop across the drain resistor and the voltage drop across the source resistor.

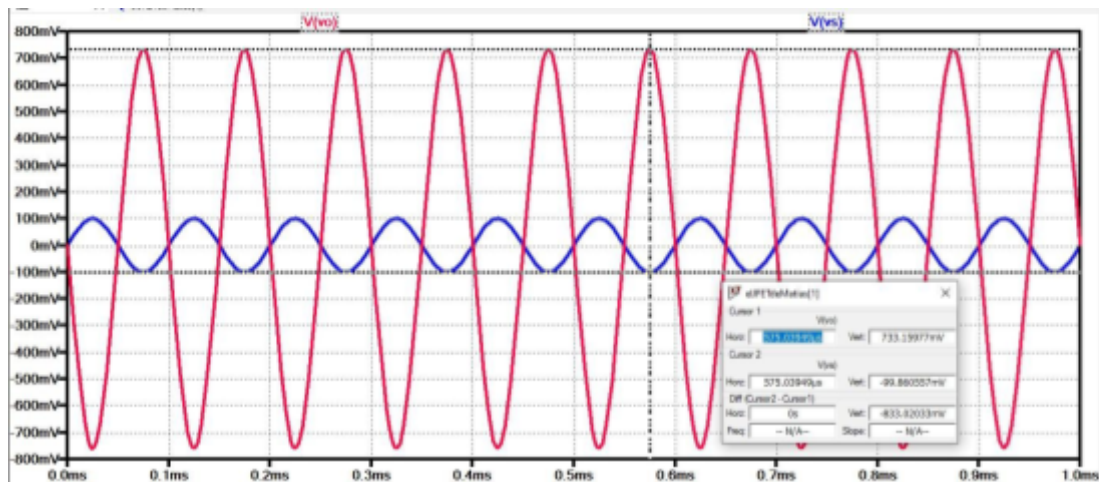


Gráfico 41

Gain obtained with  $V_s = -99.86\text{mV}$  and  $V_o = 733.16\text{mV}$ .  $AVS = 7.34\text{ V/V}$ .

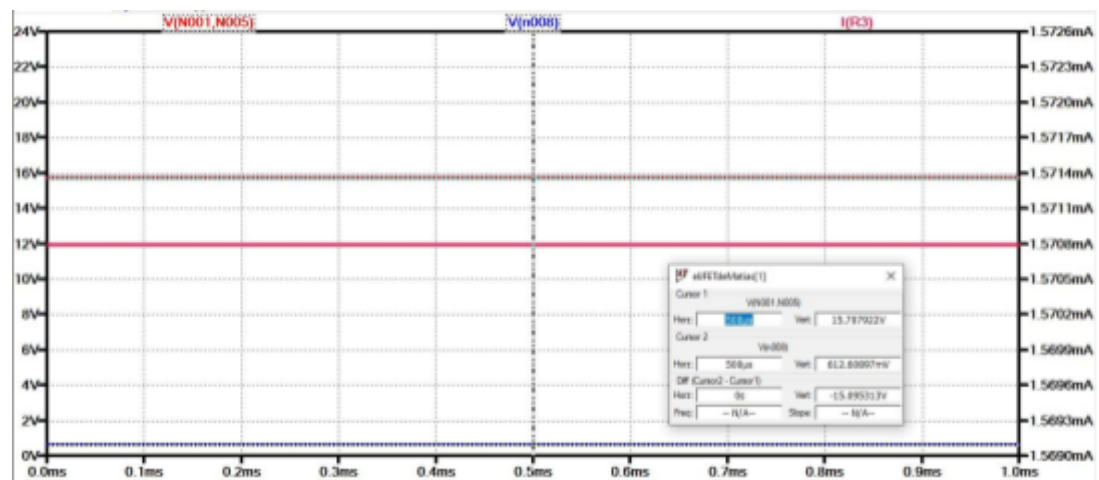
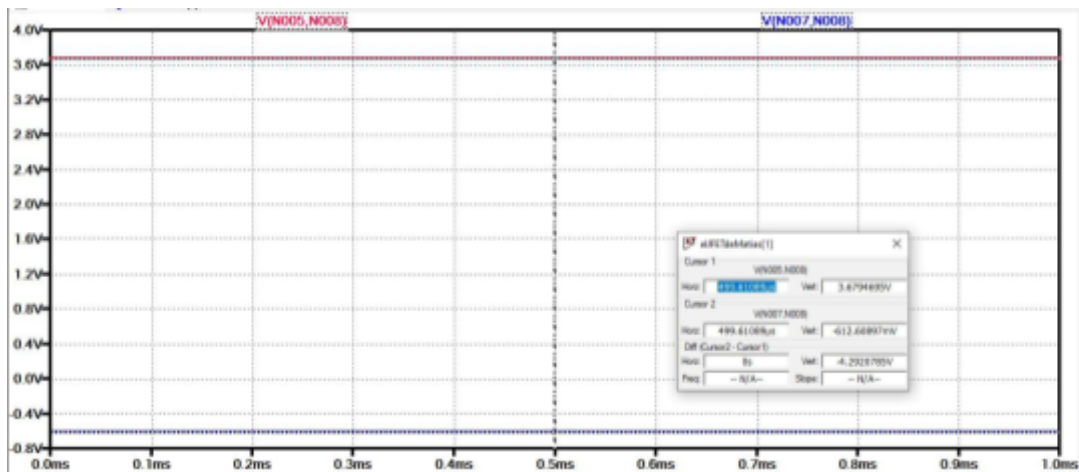


Gráfico 42

Working current ( $I_d = 1.57\text{ mA}$ ), and voltage drops ( $V_{RD} = 15.7\text{ V}$ ) ( $V_{RS} = 0.61\text{ V}$ ).

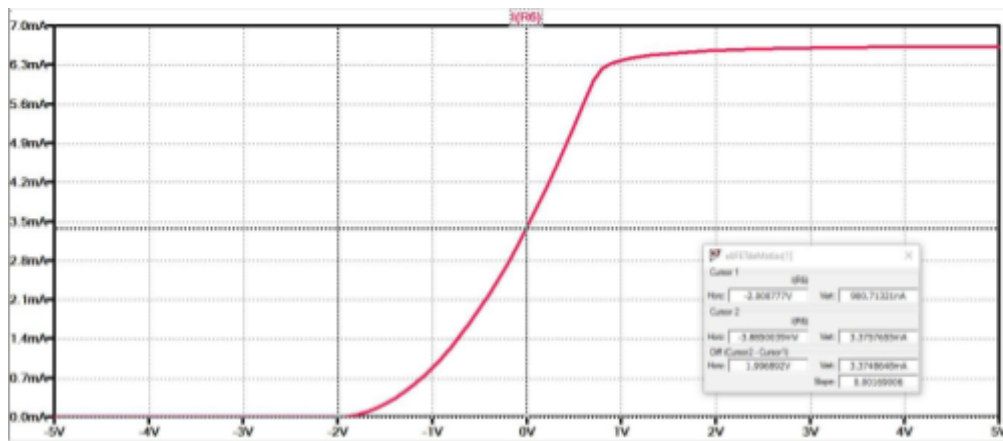
For the voltage drops between gate and source and drain and source we use graph 43.





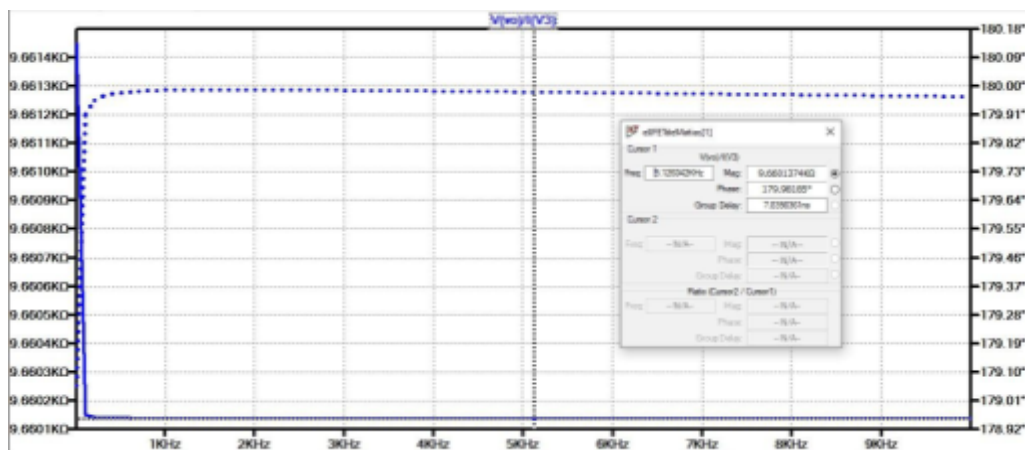
$V_{gs}$  (-0.6V), and  $V_{ds}$  (3.68V) of the SC configuration.

Figure 44 shows the values of  $D_{ss}$  and  $V_p$ .



Construction characteristics,  $I_{dss}$  (3.37 mA) and  $V_p$  (-2 V).

We also simulated the corresponding ROA value in chart 45.



Simulated ROA.(9k66 $\Omega$ ).

### 3.4 Measurements

Once the simulation has been carried out based on the values calculated on paper and pencil, all that remains is to verify the operation with real components.

A breadboard was used to position all the elements used and the resistors chosen were 1/4W and 5%.

The first thing that was checked was the voltage gain in graphs 46 and 47, which was one of the main specifications imposed by the exercise.



Gráfico 46

*Input signal measurement [99.6 mVpp]*



Gráfico 47

*Measurement of the output signal [704 mVpp].*

$$|AVS| = \frac{V_o}{V_s} = 99^{704}_{99.6} = 7.1$$

Since a fairly low amplitude was chosen for the signal generator, the presence of noise noticeably affected the input signal, and cursors were used to obtain a more precise value.

The corresponding numbers for the static working point and some intermediate values used in the previous calculations were also verified.

Using a multimeter, the voltage drop across the drain and source resistors was measured, and the operating current  $I_D$  was verified by an average, as was done for the BJT.

$$V_{RD} = 9.14 \text{ V} \Rightarrow I_{DQ} = \frac{V_{RD}}{R_D} = \frac{9.14 \text{ V}}{12 \text{ k}\Omega} = 761.6 \text{ }\mu\text{A}$$

$$V_{R_{SO}} = 3.86 \text{ V} \Rightarrow I_{DQ} = \frac{V_{R_{SO}}}{R_{SO}} = \frac{3.86 \text{ V}}{5600} = 689.28 \text{ }\mu\text{A}$$

In this way, we were able to verify  $I_D = \frac{V_{RD}}{R_D} = \frac{9.14 \text{ V}}{12 \text{ k}\Omega} = 761.6 \text{ }\mu\text{A}$  and  $I_D = \frac{V_{R_{SO}}}{R_{SO}} = \frac{3.86 \text{ V}}{5600} = 689.28 \text{ }\mu\text{A}$  of the 44 points with the values

measured.

Also  $V_{DSQ} = V_{DD} - I_D (R_S + R_D) = 20 \text{ V} - 772 \text{ }\mu\text{A} \cdot (5600 \text{ }\Omega + 12000 \text{ }\Omega) = 16.41 \text{ V}$

maximum in graph 48 and the equivalent of the thevenin model in graph 49.



Gráfico 48



Gráfico 49

Measurement of the  $I_{DSS}$  current (16.4 mA) and the voltage drop across  $R_2$  (1.28 V).

To measure  $I_{DSS}$ , connect the Gate to the Source (bridge these terminals) and then place the multimeter in ammeter mode across the output loop, ensuring the probes are in series with a relatively small Drain resistor (it doesn't need to be the same as the  $R_{Drain}$  of the amplifier's JFET). Once the loop is closed, the value displayed on the screen is  $I_{DSS}$ .

After all the calculations and measurements made up to this point, we studied the cutoff limits and ohmic region limits, as well as the maximum excursion of  $V_o$ . This was done by defining the following limits as follows:

Cut-off limit  $\rightarrow I_{DQ}$  \*  $R_{Din} = b$  where  $R_{Din} = R_D // R_L$

Limit and asymptote by ohmic region by Ohmic  $|V_P| = 3 \rightarrow I_{DQ}$  Define limit parabola of equation  $I_D = K (V_{GS} - V_{THRESHOLD})^2$

Where, as we recall, we have calculated  $b = 5 V$  and  $I_{DQ} = 750 \mu A$ . We see that because  $b = 5 V$  and by ohmic region  $I_{DQ}$  is left over  $= 750 \mu A$ .  $V_{DSQ} = 4.275 V$  graphically  $= 7.475 V$  cutting the

curve with the limiting parabola.

We observed the empirical values of  $b = 5.146 V$  and  $I_{DQ} = 772 \mu A$ . We see that by cutting  $b = 5.146 V$  and by ohmic region  $I_{DQ}$  is left over  $= 772 \mu A$ .  $V_{DSQ} = 4.01 V$  graphically  $= 7.81 V$  cutting the curve

with the limit parabola.

We see that there is almost no difference in the cutoff limit, as it shifts by about 0.146 V, as does the ohmic region limit, which shifts by almost 0.265 V.

Given that in both cases it happens that  $I_{DQ} = 772 \mu A$ . It turns out that the transistor is region-limited.

Ohmic by the  $V_{OMAX}$  that the  $= 4.01 V$  excursion  $b$  symmetrical  $> a$ . It's from  $V_{OMAX} = 4.275 V$  with data

calculated and  $I_{DQ} = 772 \mu A$  with empirical data.

We now observe the static (RCE) and dynamic (RCD) load lines as well as the ohmic region limitations obtained by the graphical method seen in class in graph 50.

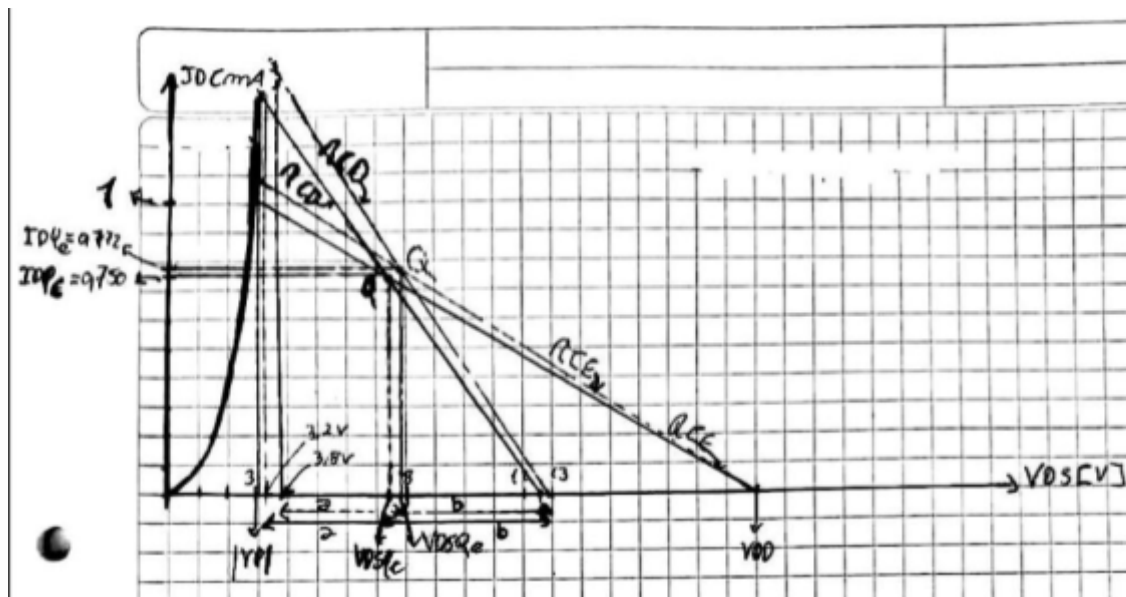


Gráfico 50

RCE and RCD for calculated data in solid line and for empirical data in dotted line.

$I_{dq}$  = Empirical drain-source current,  $V_{dsq}$  = Empirical drain-source voltage

$I_{dq}$  = Calculated  $I_{dq}$  current,  $V_{dsq}$  = Calculated Drain-Source

voltage **Calculated data:**

Si  $I_{dq} = 0 \text{ A} \Rightarrow V_{dsq} = V_{DD} = 20 \text{ V}$ , y si  $V_{dsq} = 0 \text{ V} \Rightarrow I_{dq} = R_{SO}^{V_{DD}} + R_D = 1.1976 \text{ mA}$  Por otro lado, el

punto Q queda  $Q = \{V_{dsq}, I_{dq}\}$  donde  $V_{dsq} = 7.475 \text{ V}$  y  $I_{dq} = 750 \text{ uA}$

Además  $I_{dss} = 15 \text{ mA}$  impone una restricción en  $V_{dsq}$  de  $V_{dsq} \leq V_{DS_{max}}$  y  $R_{CD} \leq R_{CE}$ . También  $V_{DS_{max}} = 1.8731 \text{ V}$  la asíntota en el diseño gráfico

The limit of the parabola is at  $|V_p| = 3$

**Empirical data:**

Si  $I_{dq} = 0 \text{ A} \Rightarrow V_{dsq} = V_{DD} = 20 \text{ V}$ , y si  $V_{dsq} = 0 \text{ V} \Rightarrow I_{dq} = R_{SO}^{V_{DD}} + R_D = 1.1976 \text{ mA}$  Por otro lado, el

punto Q queda  $Q = \{V_{dsq}, I_{dq}\}$  donde  $V_{dsq} = 7.81 \text{ V}$  y  $I_{dq} = 772 \text{ uA}$

Además  $I_{dss} = 15 \text{ mA}$  impone una restricción en  $V_{dsq}$  de  $V_{dsq} \leq V_{DS_{max}}$  y  $R_{CD} \leq R_{CE}$ . También  $V_{DS_{max}} = 1.9513 \text{ V}$  la asíntota en el diseño gráfico

The limit of the parabola is at  $|V_p| = 3 \text{ V}$

As mentioned in the design section, we used various configurations to meet the group's specifications, so we also tested these circuits in practice, focusing on the best result.

We experimentally obtained a gain slightly greater than 7 V/V as can be seen in graph 51.

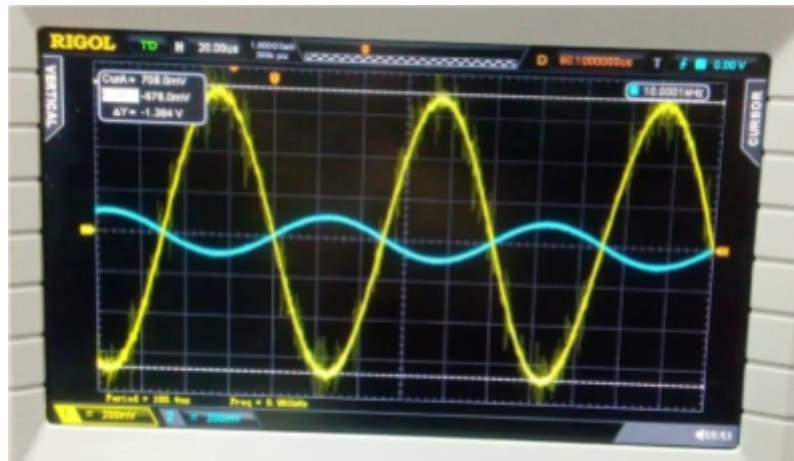


Gráfico 51

AVS of the SC configuration. (7.1 V/V).

Figures 52 and 53 show the voltage drops corresponding to the drain and source resistors. It is important to mention that the latter was modified in the experiment because there was a slight difference in the expected current; therefore, a 120-ohm resistor was used after fine-tuning.



Gráfico 52



Gráfico 53

Medición de las caídas de tensión sobre la resistencia de drain (14.92 V) y sobre la de source (0.18V).

Gráfico 49

$$V_{RD} = 14.92 \text{ V} \Rightarrow I_D = \frac{V_{RD}}{R_D} = \frac{14.92 \text{ V}}{10^3 \Omega} = 1.49 \text{ mA}$$

$$V_{RSO} = 0.18 \text{ V} \Rightarrow I_D = \frac{V_{RSO}}{R_{SO}} = \frac{0.18 \text{ V}}{120 \Omega} = 1.53 \text{ mA}$$



We also measure the value of  $I_D$  at  $V_{GS} = 1.49 \text{ mA} + 2.54 \text{ mA} = 1.53 \text{ mA}$  and of  $V_{DS} = 1.51 \text{ mA}$  of the circuit.



Gráfico 54



Gráfico 55

$V_{GS} (-0.654 \text{ V})$  and  $V_{DS} (-3.94 \text{ V})$  for the SC configuration.

To obtain the ROA value, we simply connect a potentiometer to the circuit output at zero and analyze the amplitude of the output signal; then we gradually increase the resistance value until we obtain half of that amplitude. Finally, we measure the potentiometer (Figure 56) when that condition is met and obtain the ROA.



Gráfico 56

ROA measured for the SC. (9k890h).

To ROA the RIA (calculated) simply = 8 we measured  $k8\Omega$  ROA the resistance (simulated) connected =  $9k66\Omega$  at gate ROA of the (measured) transistor, =  $9k89\Omega$  with

You can see it in chart 57.



Gráfico 57

*RIA measurement for all SC circuits. (4650h).*

$$RIA(\text{calculated}) = 470\Omega$$

$$RIA(\text{simulated}) = 470\Omega$$

$$RIA(\text{measured}) = 465\Omega$$

## 4. Conclusions

### TBJ

Table I shows the calculated, simulated, and measured DC and signal parameter values for the bipolar junction transistor.

|              | Calculated | Simulated | Measured |
|--------------|------------|-----------|----------|
|              | 3.114      | 3.17      | 3.18     |
| $I_c$ [mA]   | 3.9        | 3.6       | 3.28     |
| $V_{ce}$ [V] | 0.115      | 0.124     | 0.127    |
| $g_m$ [S]    | 128        | 117.9     | 127.72   |
| $AVS$ [V/V]  | ~1         | 0.997     | 0.987    |
| $A_i$ [A/A]  | 8.5        | 9.29      | 8,4      |



|                |         |         |        |                |
|----------------|---------|---------|--------|----------------|
| $RIA [\Omega]$ | 4300.47 | 4257.32 | 4737,9 | $ROA [\Omega]$ |
|----------------|---------|---------|--------|----------------|

Tabla I

## JFET

The process was repeated for the JFET, and the values of interest are presented in Table II for calculations, simulation, and practical application. Considering all the circuits analyzed, we chose the best case for the final comparison (common source).

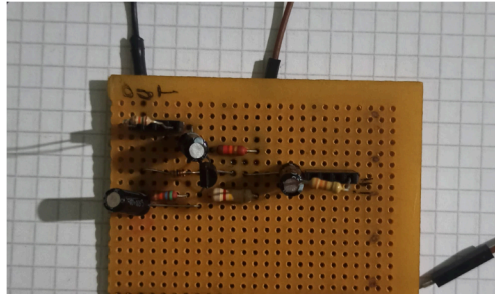
|                | Calculated | Simulated | Measured | $ROA [k\Omega]$ |
|----------------|------------|-----------|----------|-----------------|
|                | 1.51       | 1.57      | 1.51     |                 |
| $I_d [\mu A]$  | -0.59      | -0.6      | -0.654   |                 |
| $V_{gs} [V]$   | 4.415      | 3.68      | 3.94     |                 |
| $V_{ds} [V]$   | 2.19       | 2         | 2.02     |                 |
| $g_m [mS]$     | 3          | 3         | 3.1      |                 |
| $I_{dss} [mA]$ | 7.31       | 7.34      | 7.1      |                 |
| $AVS [V/V]$    | 470        | 470       | 465      |                 |
| $RIA [\Omega]$ | 8.8        | 9.66      | 9.89     |                 |

Tabla II

Analyzing both the TBJ and JFET tables, we see that the dispersion in the parameters among the three cases—"calculated," "simulated," and "measured"—is due to several factors, including:

- Noise is introduced during measurements due to the use of a breadboard circuit and cables that are not all in good condition, as well as the measuring equipment and its probes, and the temperature of the amplifier devices. When working with weak signals, any noise, however small, enters the circuit and affects the measurements.

-Considering the noise and handling problems that come with using a protoboard, we opted to try making a PCB for the amplifier with a FET in the common gate configuration; an image of the result is attached.



- The use of certain parameters in the transistor models during simulation causes some slight changes in the simulated parameters compared to what is desired.

- The use of approximations in the calculations and some parameters from the data sheets, as well as the use of commercial resistors instead of the exact calculated resistors, introduce some variation in the parameters.

We encountered several problems during the analysis of the JFET circuit. While all the calculations on paper made sense and met the given specifications, some values didn't match when simulated with the parameters of our actual transistor. Similarly, in the lab, the currents and voltages tested were similar, but the gain failed to exceed 5V/V. For this reason, we chose to use a completely new transistor with characteristics quite different from what we had been using and analyzed in depth the case that came closest to the expected values.

## 5. Appendix

This appendix addresses the discrepancy observed between the gains calculated through simulation and those obtained experimentally. The analysis is performed using computational tools and is based on the hypothesis that, for similar values of the dss current ( $I_{dss}$ ), there is a linear relationship between the gain ( $av_s$ ) and the ds voltage ( $V_{ds}$ ), assuming that  $I_{dss}$ ,  $r$ , and  $r_D$  remain constant. Based on the experimental values, a linear regression is performed considering  $V_{ds}$  as the independent variable and  $Av_s$  as the dependent variable.

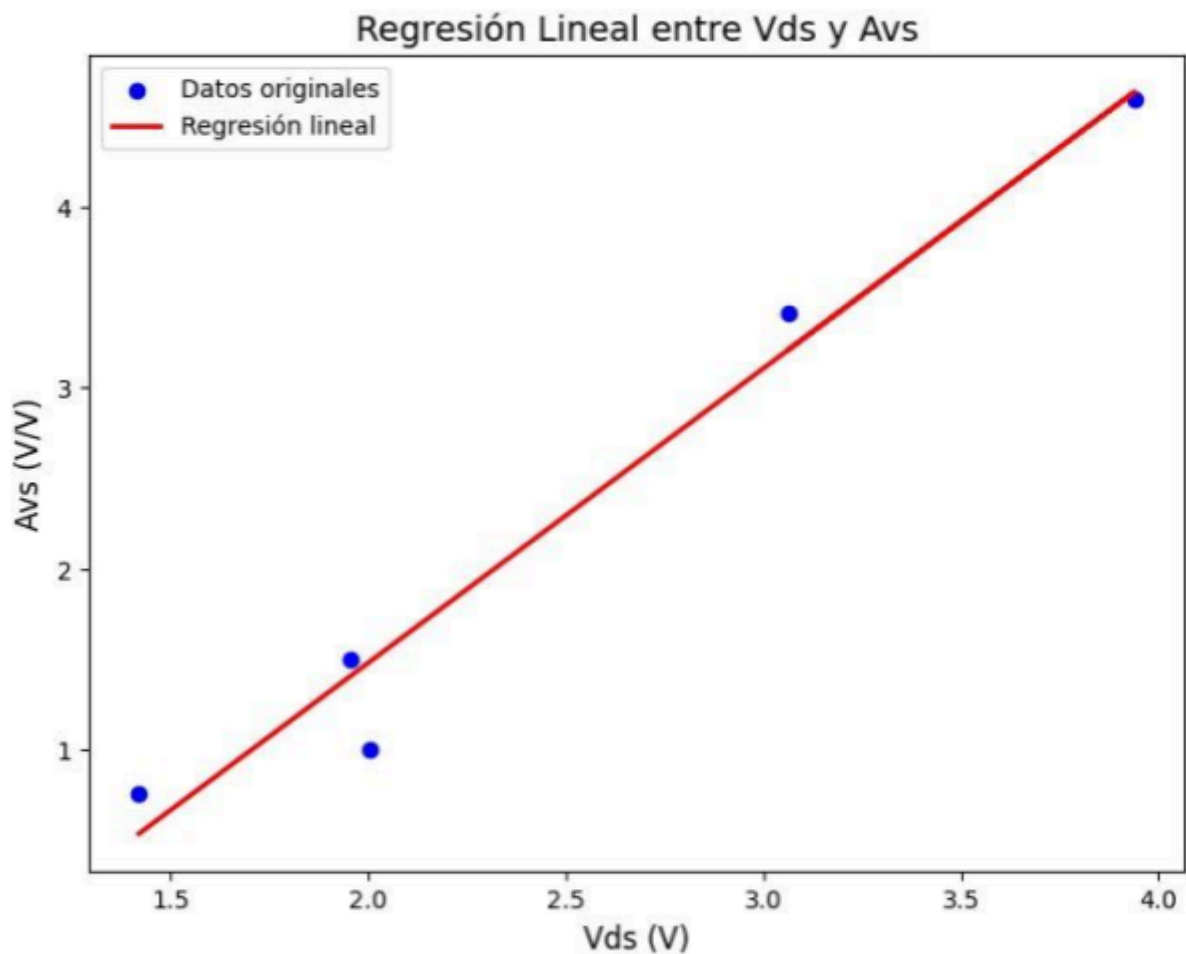


Gráfico 58

The results in Figure 58 show that increasing  $V_{ds}$  increases the gain under the aforementioned assumptions, demonstrating a positive correlation between these variables. This suggests that the only way to increase the gain would be by increasing the value of the power supply  $V_{dd}$ , since the other parameters remain constant.

However, if this analysis is omitted due to the limitations imposed by the initial hypotheses, it is possible to consider the practical behavior of the device. Reviewing the experimental data and the information provided in the practical work, it is observed that, for low values of  $I_{dq}$ , one way to increase the gain would be to increase  $r_D$ . However, this generates a greater voltage drop across  $V_{rD}$ , which, combined with the limit imposed by  $V_{dd}$  (20 V), would cause the device to move out of its operating region before reaching the desired gain.

Another viable option would be to significantly increase  $I_{dq}$ , aiming to reduce  $r_D$  to the minimum possible (keeping in mind that values very close to  $i_{dss}$  also generate instability in the circuit). Since Ohm's law maintains linear behavior in this operating range, this solution is feasible. Based on the results obtained, one of the highest gains achieved under the experimental conditions was obtained, although still below expectations. Fine adjustments to  $r_D$  allowed for an increase in  $A_{vs}$ ; however, the gain obtained remained insufficient.

Finally, an alternative and unconventional hypothesis was considered: that the acquired transistors (being the same model, but purchased from different stores and on different dates) might have a factory defect affecting the gain at the current levels handled. This behavior would not be evident at low current levels, where the gain appears to match expectations.

## 6. Literature

- PowerPoints and PDFs presented in class
- Personal notes
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